MOCCA 2025
Fundamentals of Character Animation and Motion Control

Lecture 06:

Learning-based Character Animation

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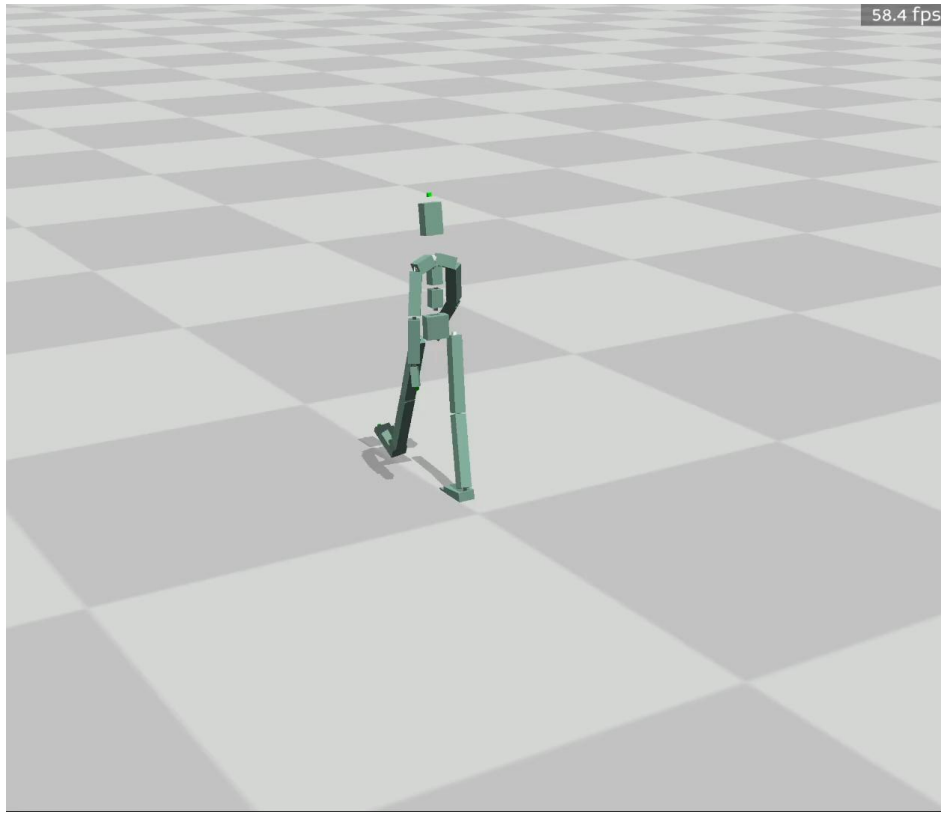
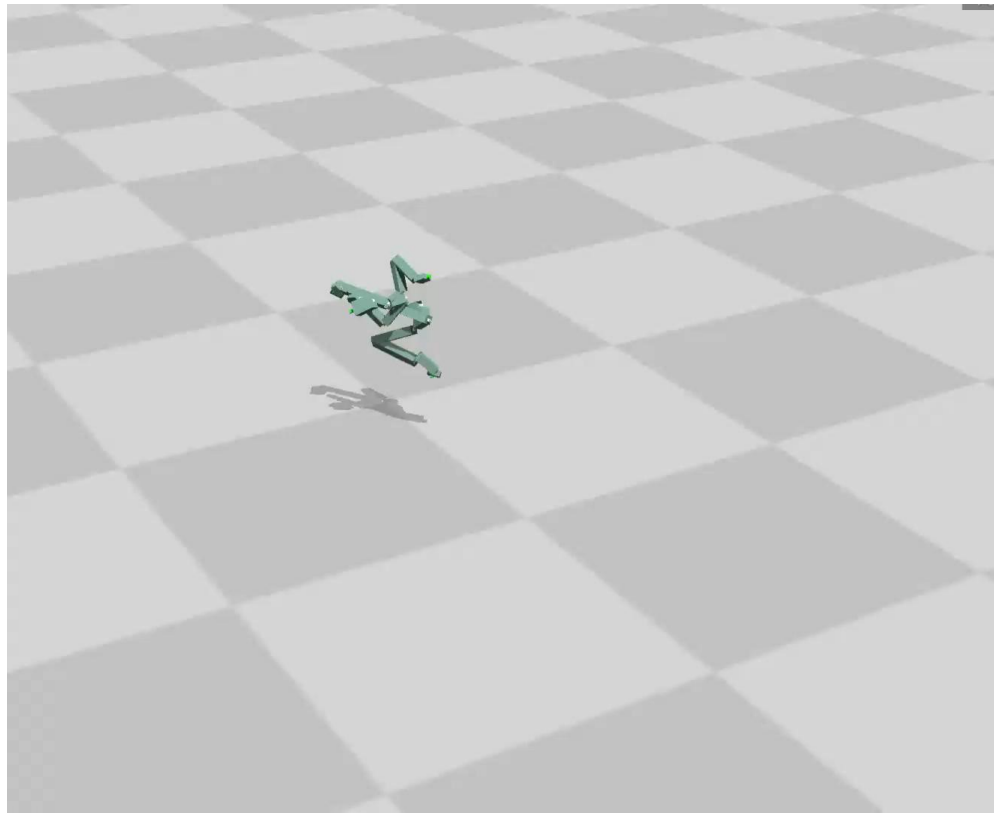
Outline

- Statistical Models of Human Motion
  - Principal Component Analysis
  - Gaussian Models
- Learning-based Models
  - Motion Models
  - Autoregressive models: PFNN
  - Generative models

Statistical Models of Human Motion

What is a natural-looking motion?





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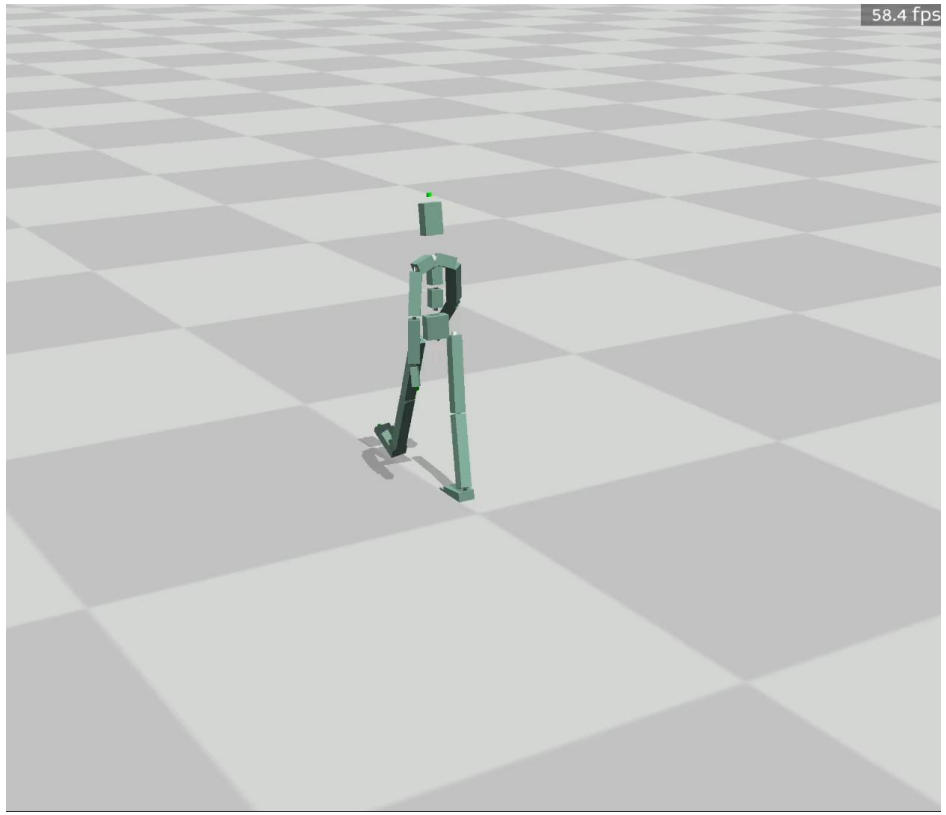

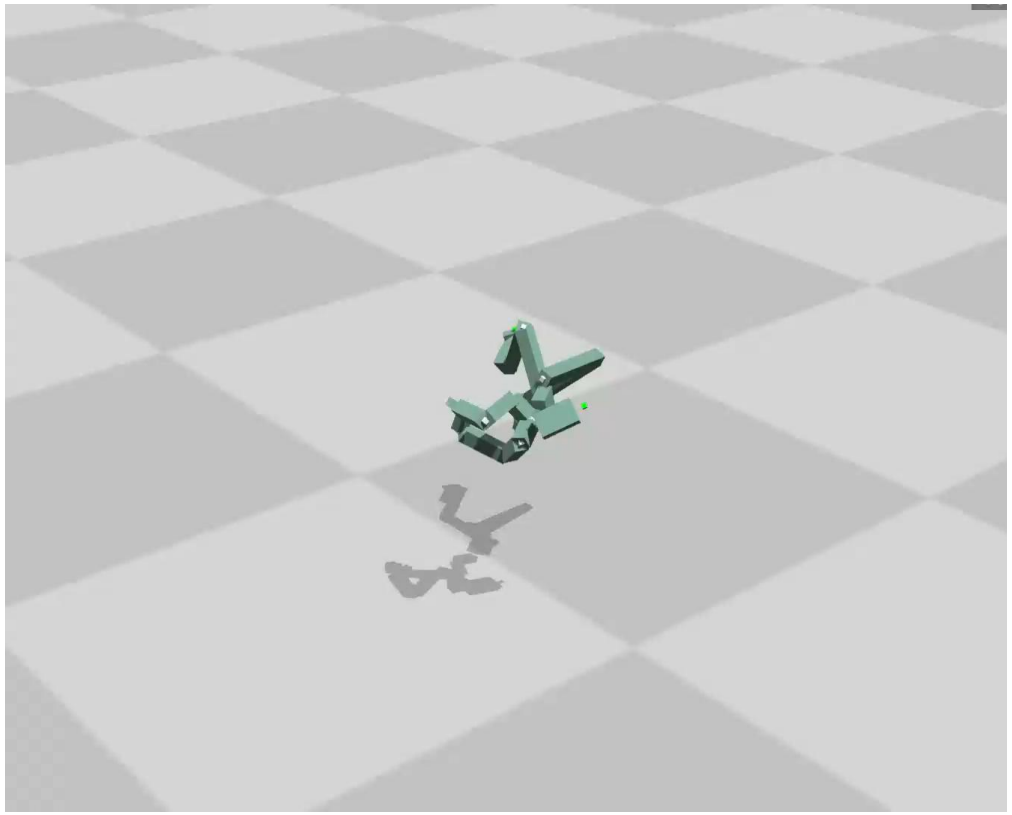

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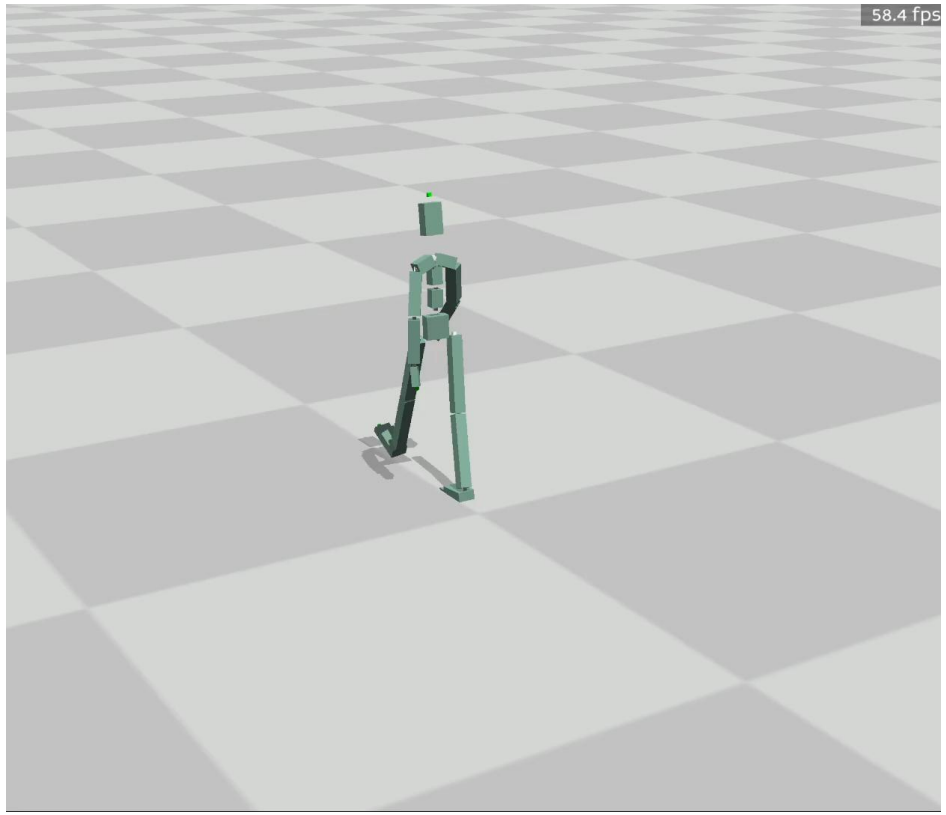
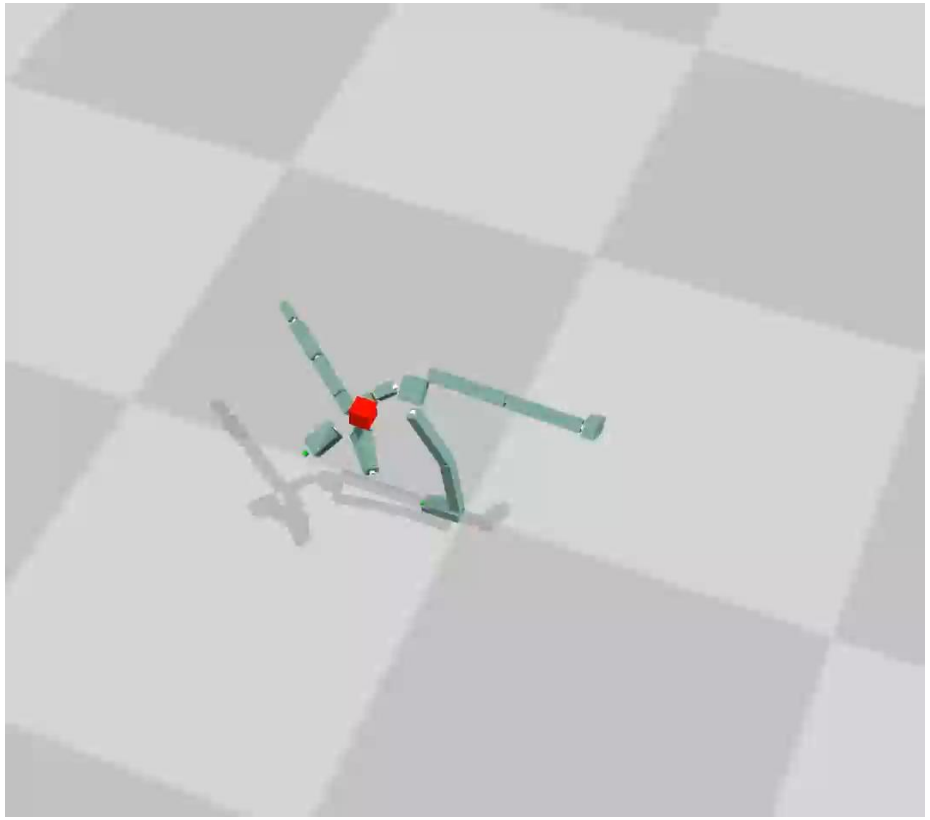
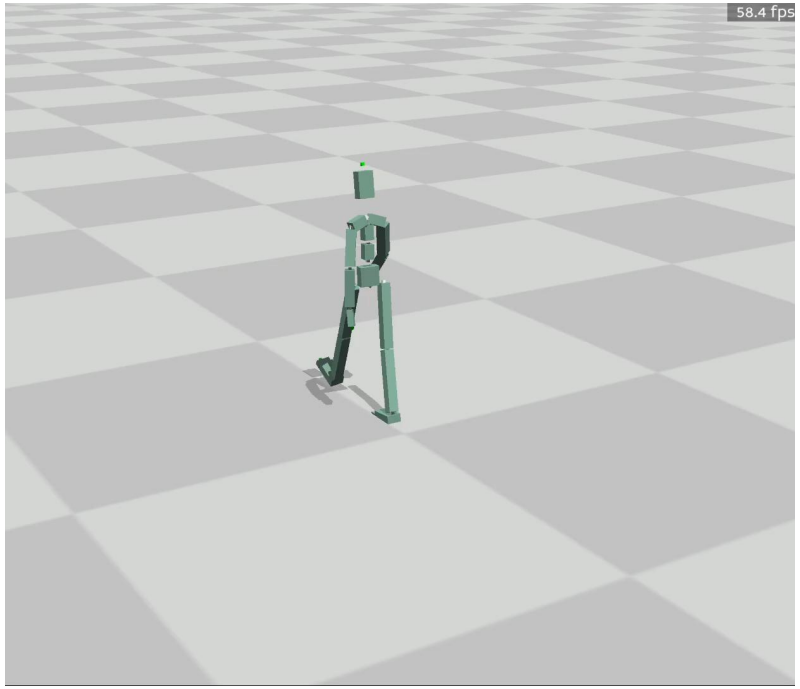


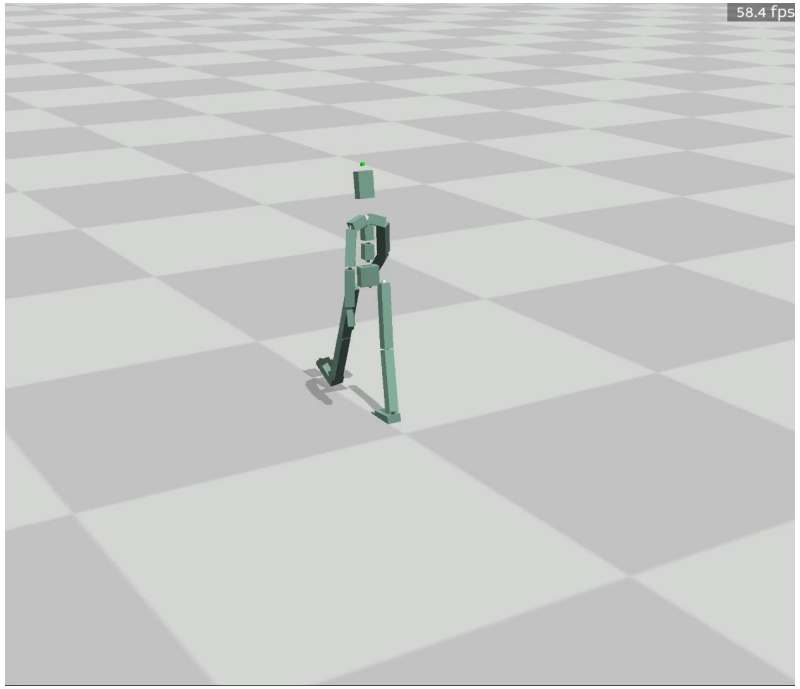


The Low-dimensionality of Human Motions

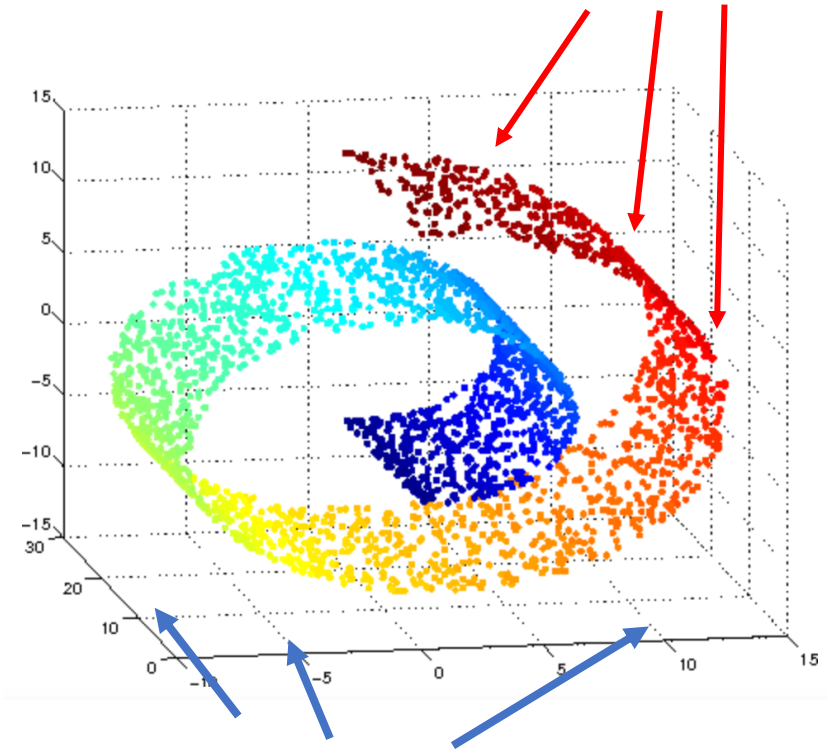


- Coordinated arm/leg movement
- Musculoskeletal structure
- Laws of physics
- ……

The Low-dimensionality of Human Motions



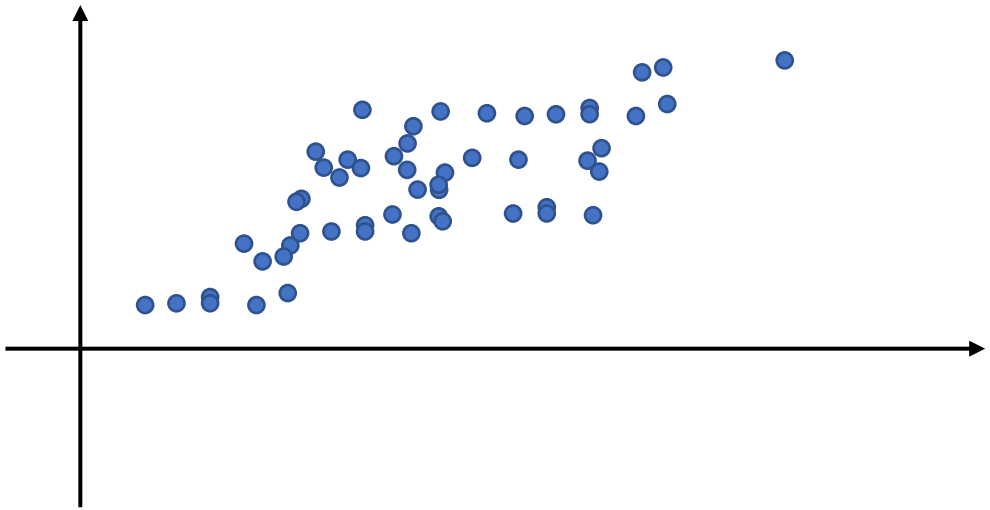
Where a natural motion locates

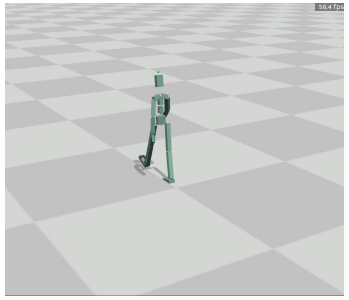


The entire “pose” space

Principal Component Analysis (PCA)

- A technique for
  - finding out the correlations among dimensions
  - dimensionality reduction





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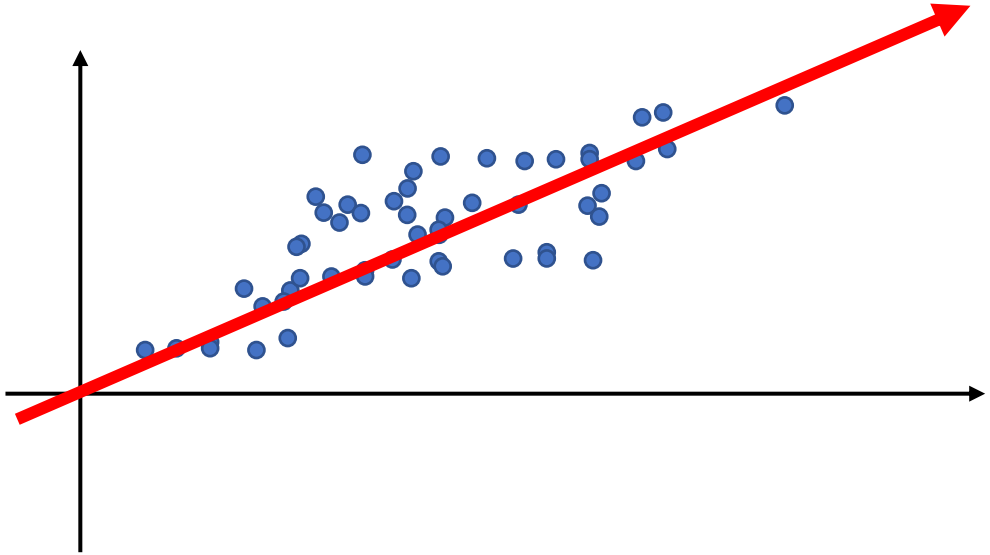
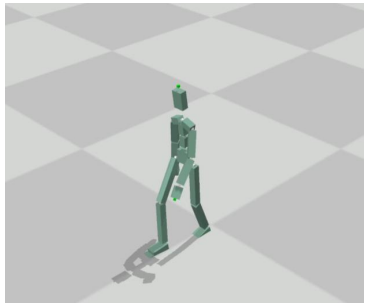


useful data



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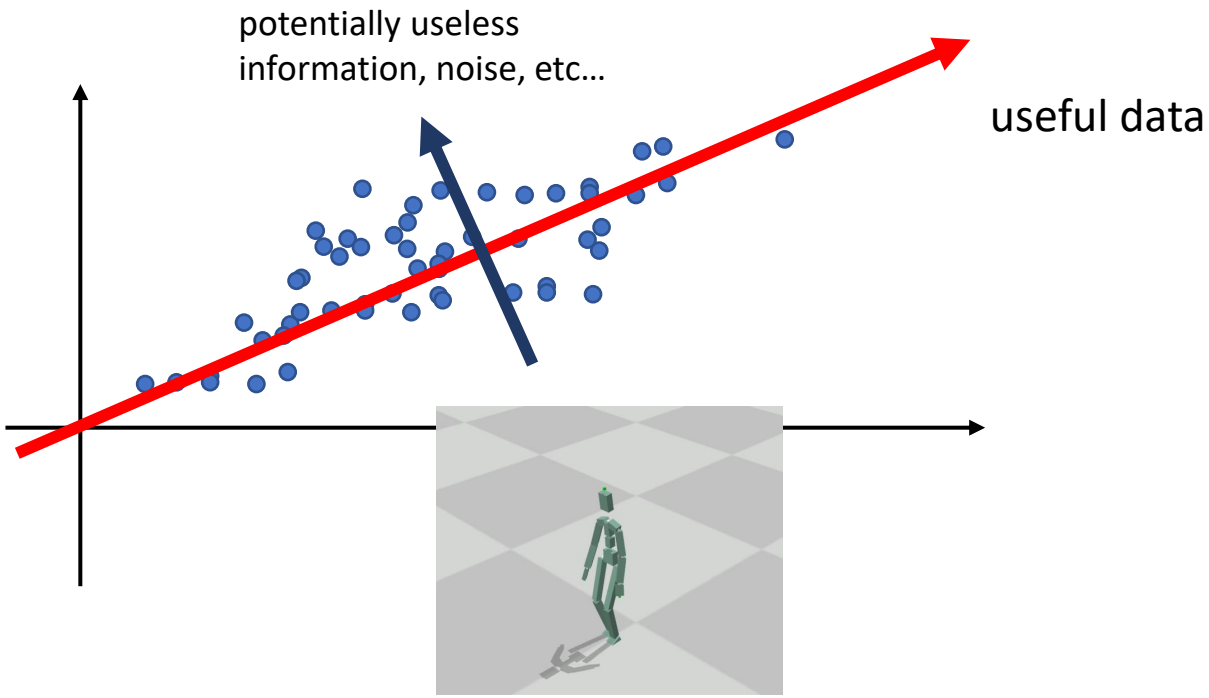
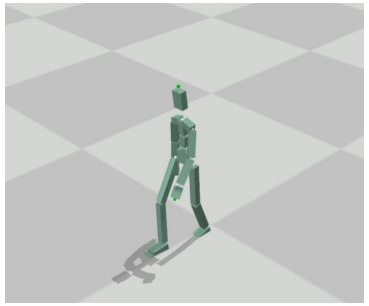
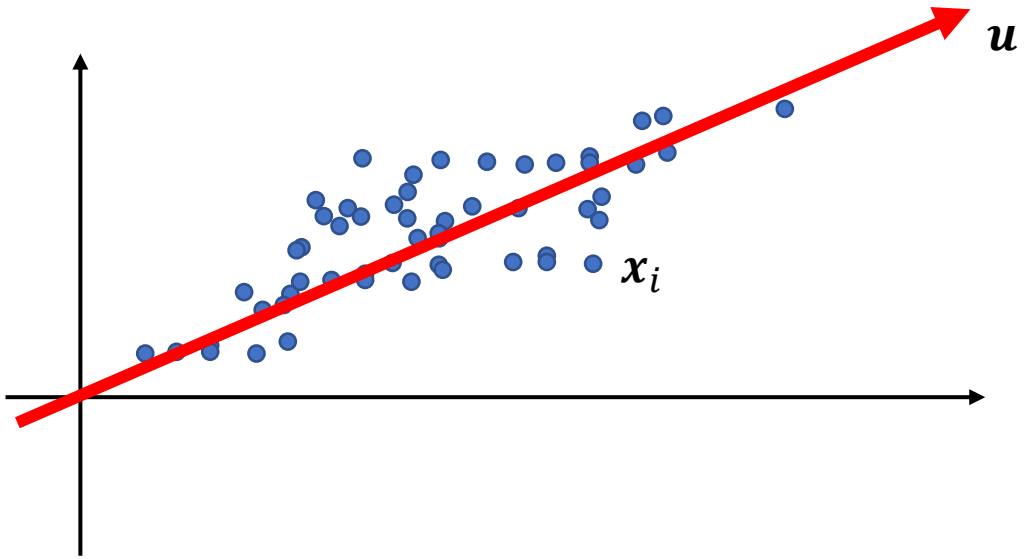
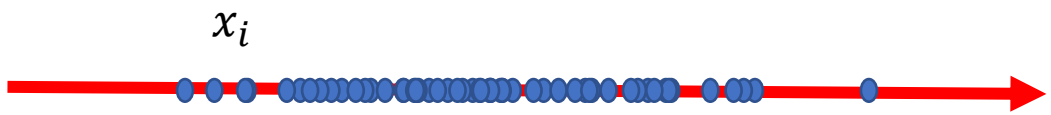




Principal Component Analysis (PCA)

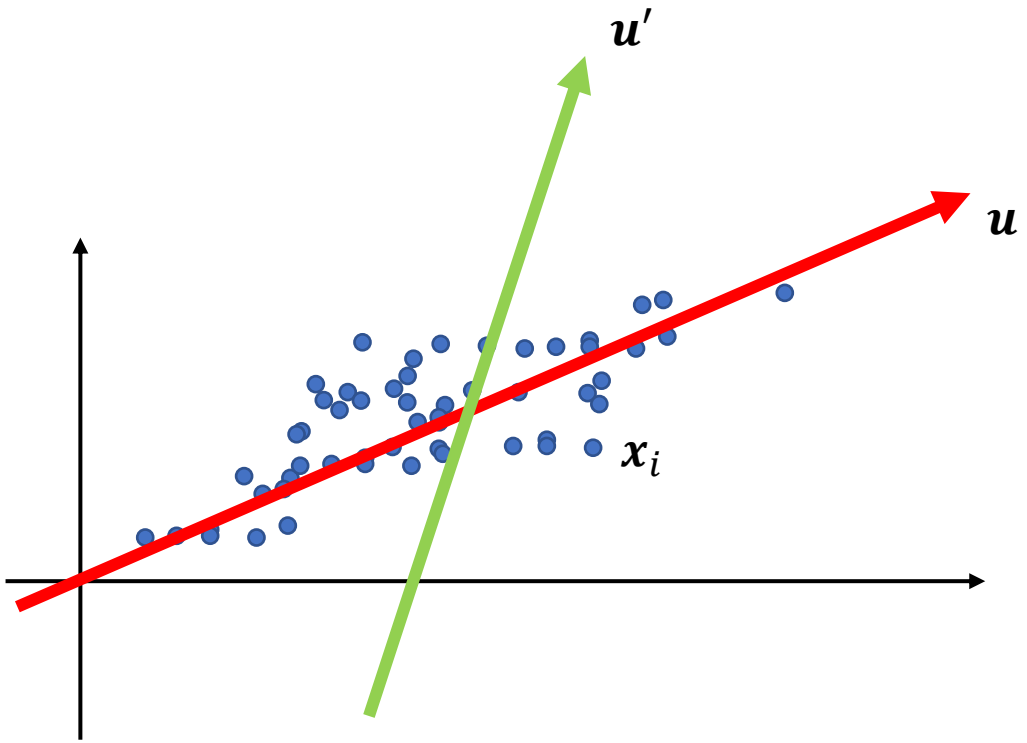
Projection of $\boldsymbol{x}_i$ on $\boldsymbol{u}$: $w_i = \boldsymbol{x}_i \cdot \boldsymbol{u}$





We can approximate $\boldsymbol{x}_i \approx w_i \boldsymbol{u}$

Principal Component Analysis (PCA)



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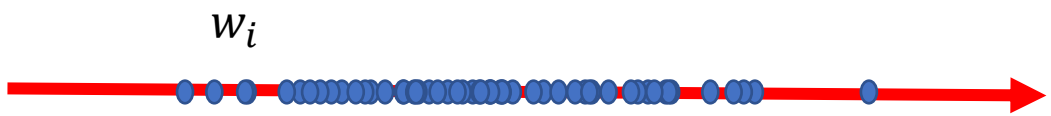
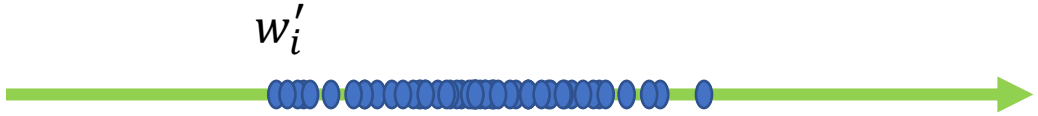


Projection of $\boldsymbol{x}_i$ on $\boldsymbol{u}'$: $w_i' = \boldsymbol{x}_i \cdot \boldsymbol{u}'$



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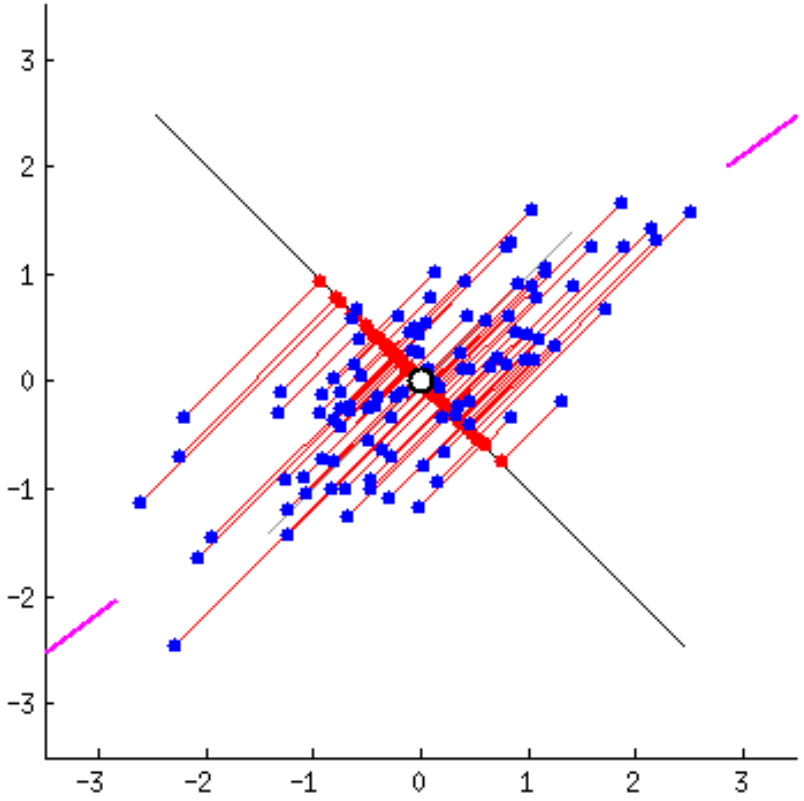

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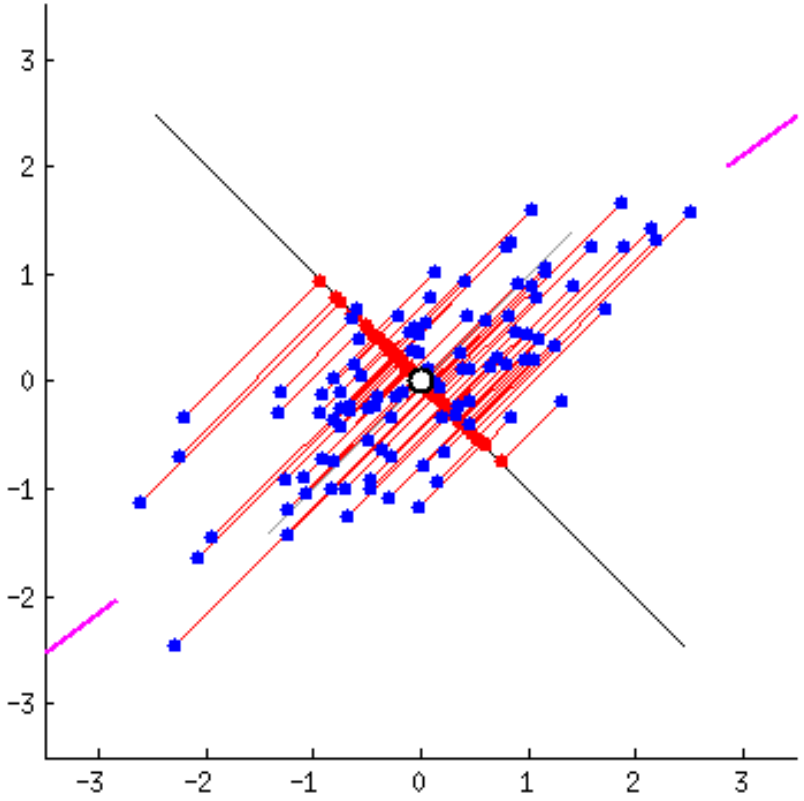

Find a direction $\boldsymbol{u}$ such that $\|\boldsymbol{u}\| = 1$, and the projections of $\{\boldsymbol{x}_i\}$ on $\boldsymbol{u}$: $w_i = \boldsymbol{x}_i \cdot \boldsymbol{u}$ have the maximal variance:

$$\frac{1}{N}\sum_i (w_i - \bar{w})^2$$

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let $X = \begin{bmatrix} (\boldsymbol{x}_0 - \overline{\boldsymbol{x}})^T \\ (\boldsymbol{x}_1 - \overline{\boldsymbol{x}})^T \\ \cdots \\ (\boldsymbol{x}_N - \overline{\boldsymbol{x}})^T \end{bmatrix}$

X

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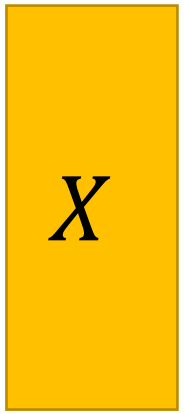
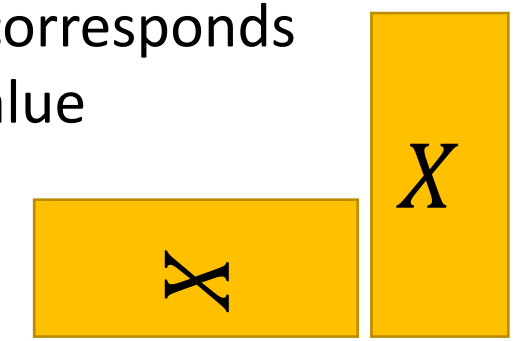


It can be proved that $\boldsymbol{u}$ is an **eigenvector** of $X^T X$ corresponds to the largest eigenvalue



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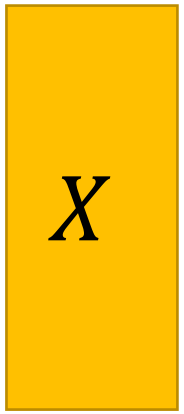

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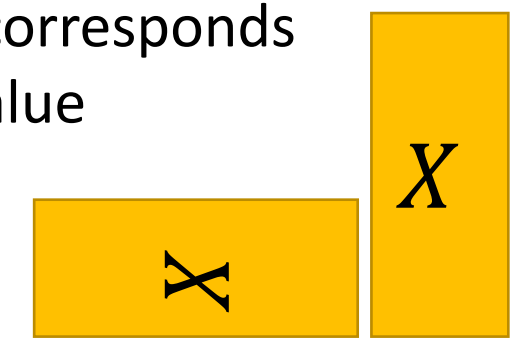


Note: we can approximate $\boldsymbol{x}_i \approx \overline{\boldsymbol{x}} + w_i \boldsymbol{u}$

PCA in a Nutshell

- Given a dataset $\{\boldsymbol{x}_i\}, \boldsymbol{x}_i \in \mathbb{R}^N$, then PCA gives

$$\boldsymbol{x}_i = \overline{\boldsymbol{x}} + \sum_{k=1}^{n} w_{i,k} \boldsymbol{u}_{\boldsymbol{k}}$$

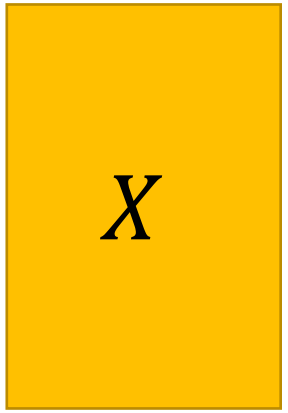
- $\boldsymbol{u}_{\boldsymbol{k}}$ is the k-th principal component
  - A direction in $\mathbb{R}^N$ along which the projection of $\{\boldsymbol{x}_i\}$ has the k-th maximal **variance**
- $w_{i,k} = (\boldsymbol{x}_i - \overline{\boldsymbol{x}}) \cdot \boldsymbol{u}_k$ is the score of $\boldsymbol{x}_i$ on $\boldsymbol{u}_k$

PCA in a Nutshell

- Given a dataset $\{\boldsymbol{x}_i\}, \boldsymbol{x}_i \in \mathbb{R}^N$, the PCA can be computed by apply **eigen decomposition** on the covariance matrix

$$\Sigma = X^T X = U \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_N^2 \end{bmatrix} U^T$$

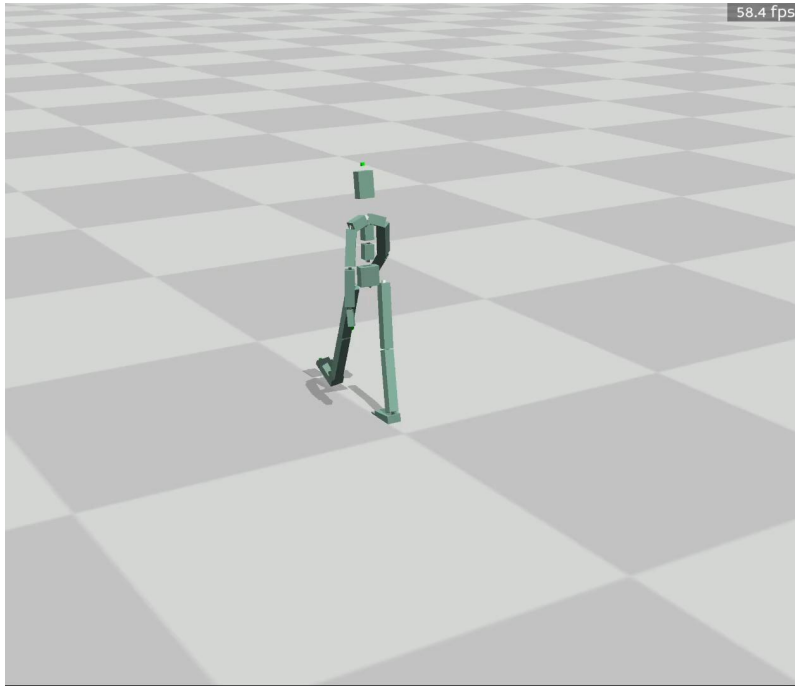
- $X = [\boldsymbol{x}_0 - \overline{\boldsymbol{x}}, \boldsymbol{x}_1 - \overline{\boldsymbol{x}}, \dots, \boldsymbol{x}_N - \overline{\boldsymbol{x}}]^T$
- $\sigma_i \geq \sigma_j \geq 0$ when $i < j$, corresponds to the Explained Variance
- $U = [\boldsymbol{u}_1, \boldsymbol{u}_2, \dots, \boldsymbol{u}_N]$



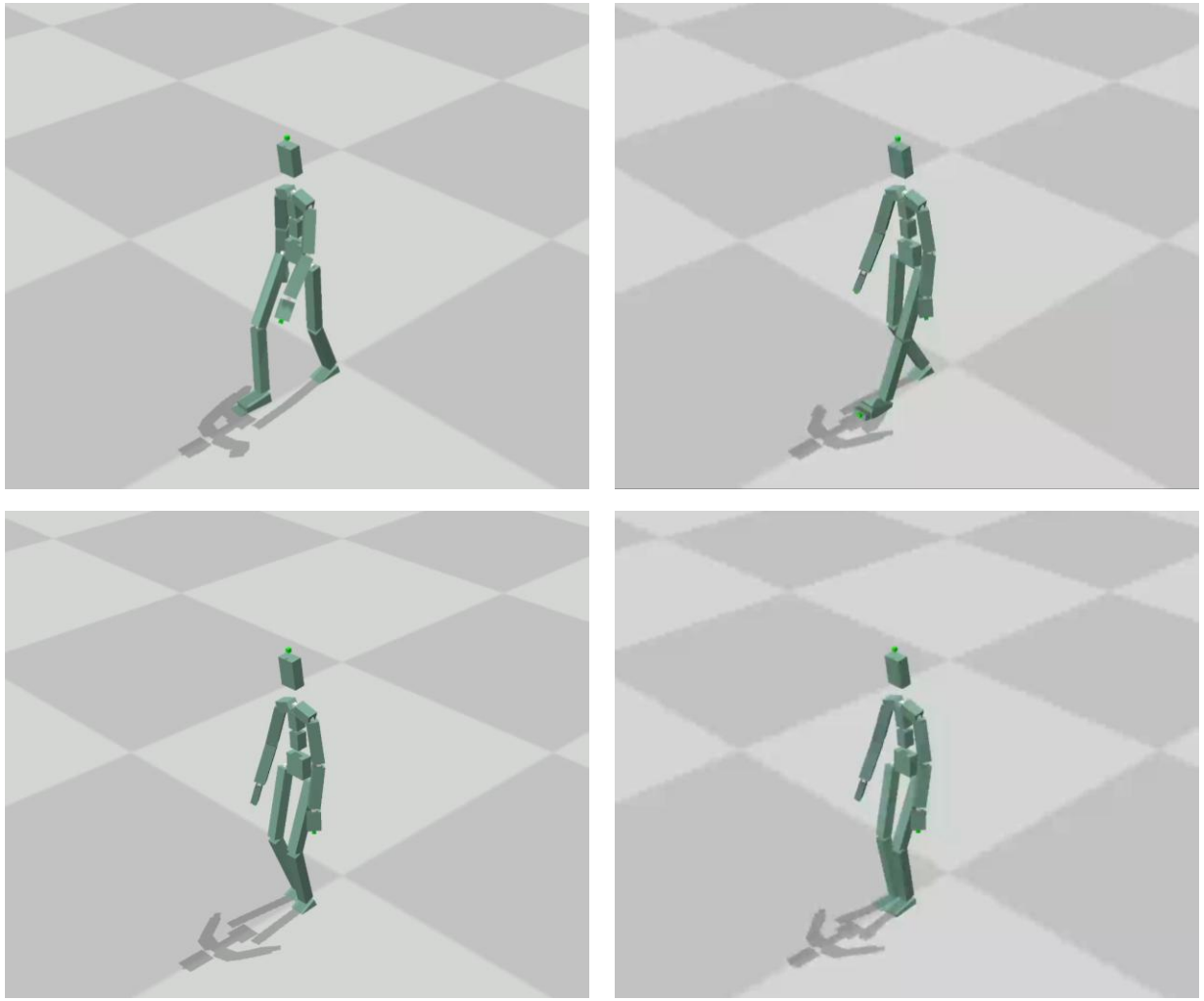
PCA Components

https://medium.com/@sebastiannorena/pca-principal-components-analysis-applied-to-images-of-faces-d2fc2c083371

PCA of Walking



$\boldsymbol{x}_i$: joint rotations



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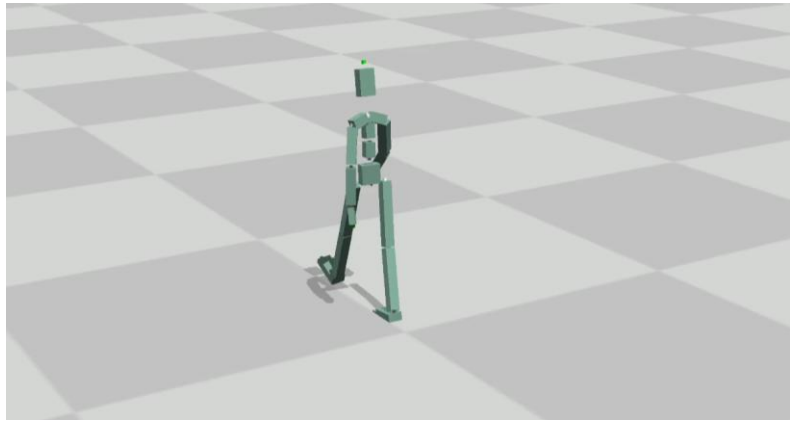
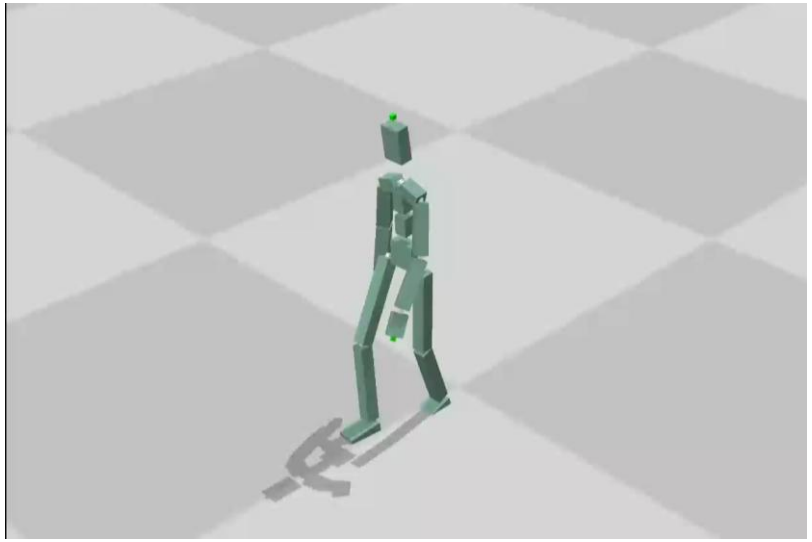
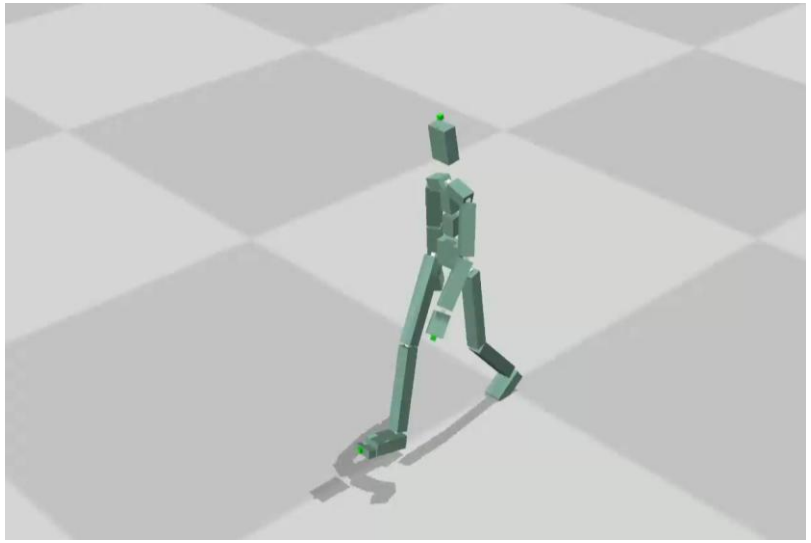
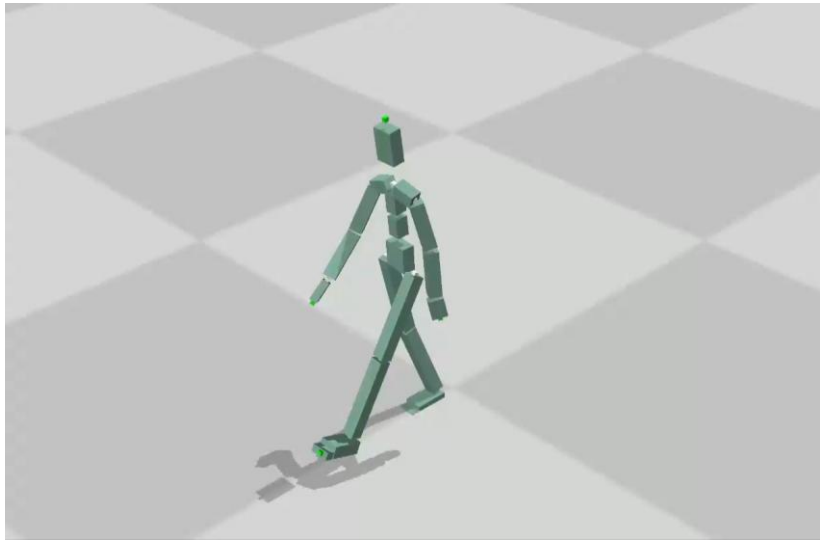
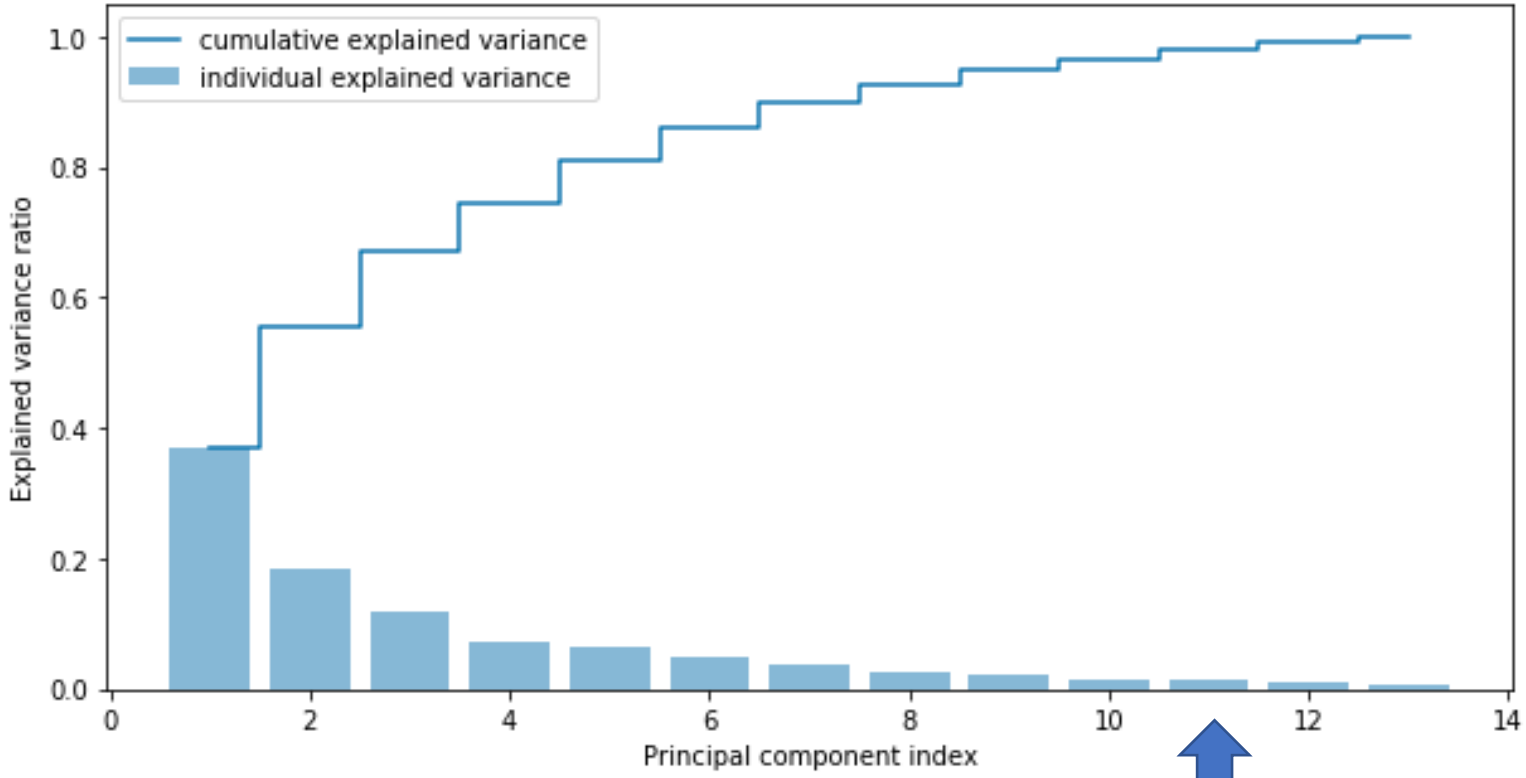






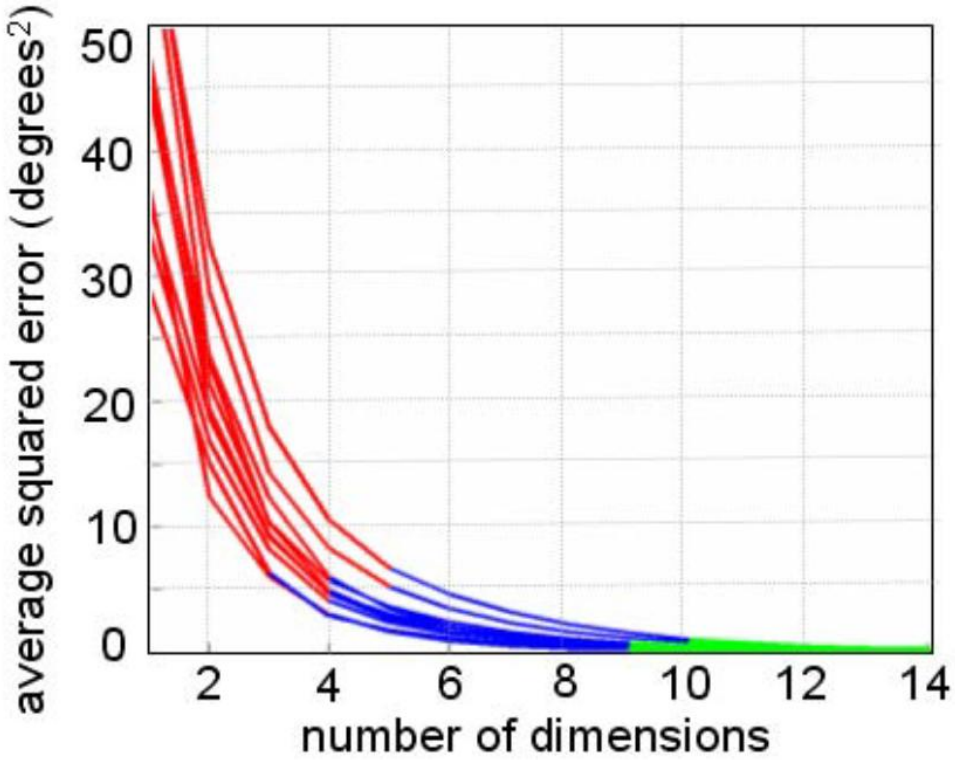


PCA Components

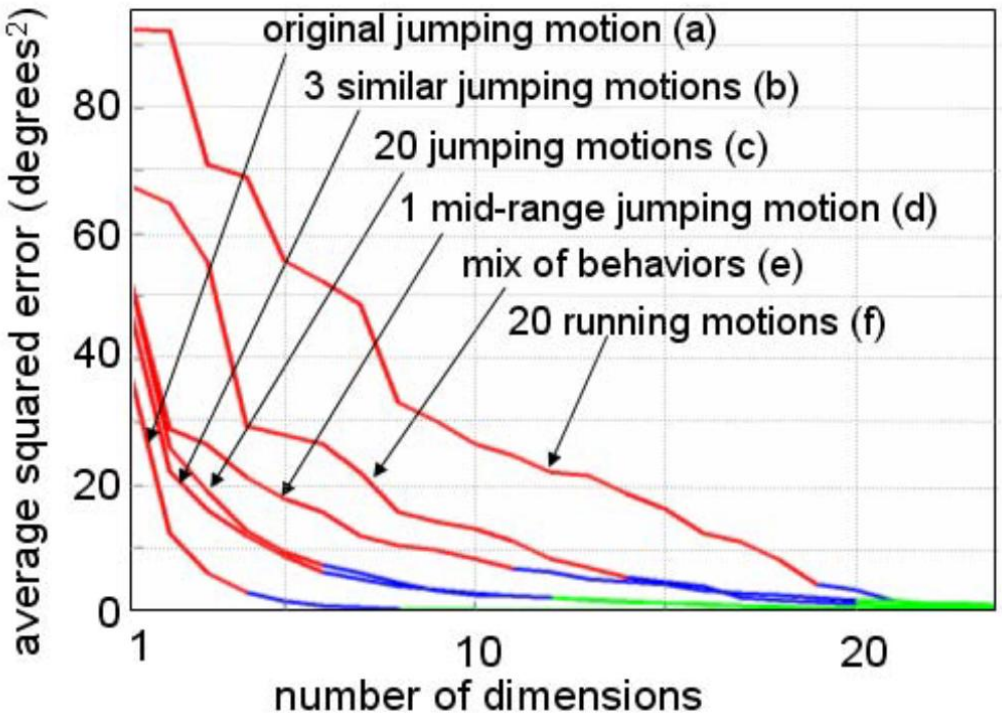


Choose n principal components that explains enough (e.g. 95%) of the variance

PCA Components



Reconstruction error of motions in the same dataset



Reconstruction error of motions in a different dataset

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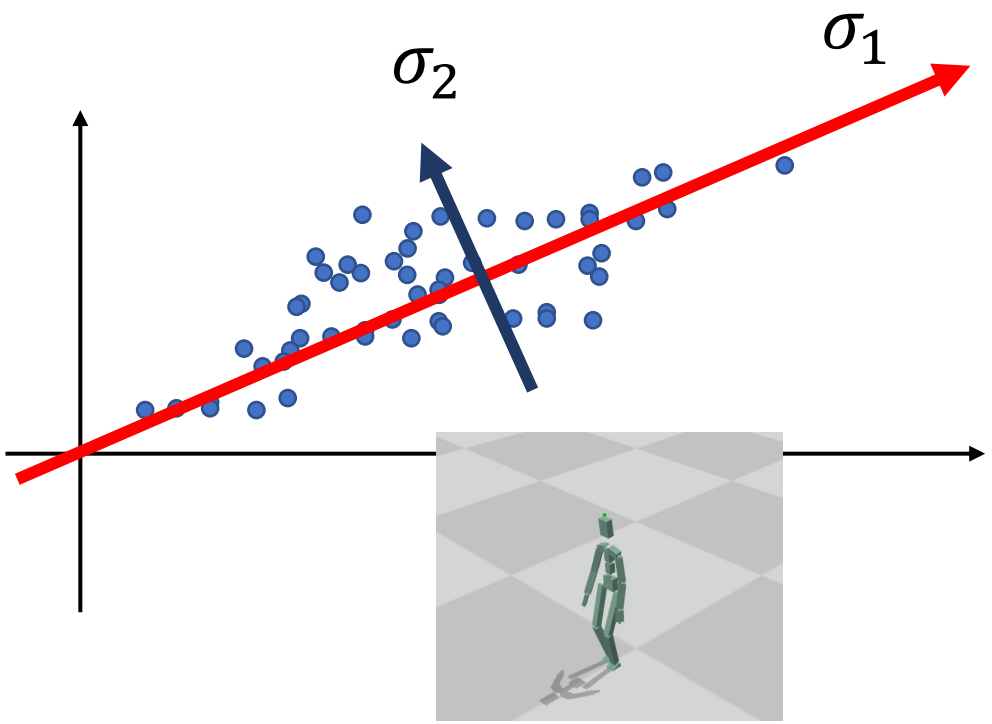

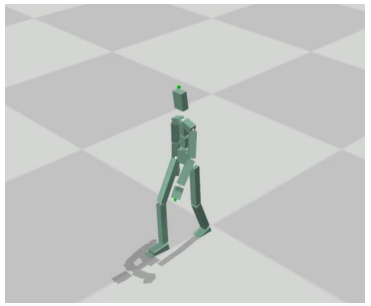

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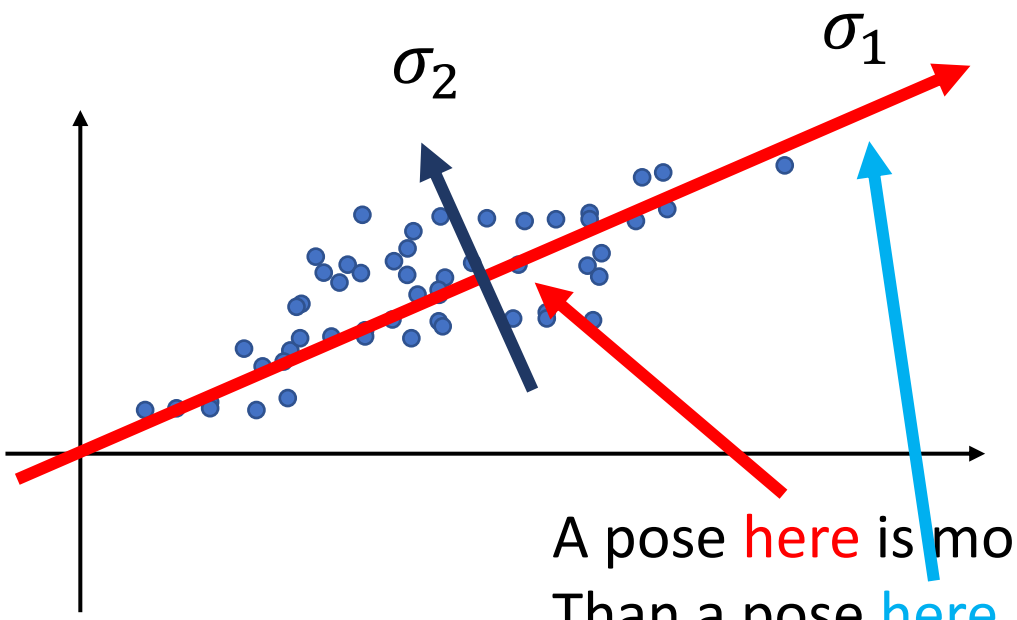


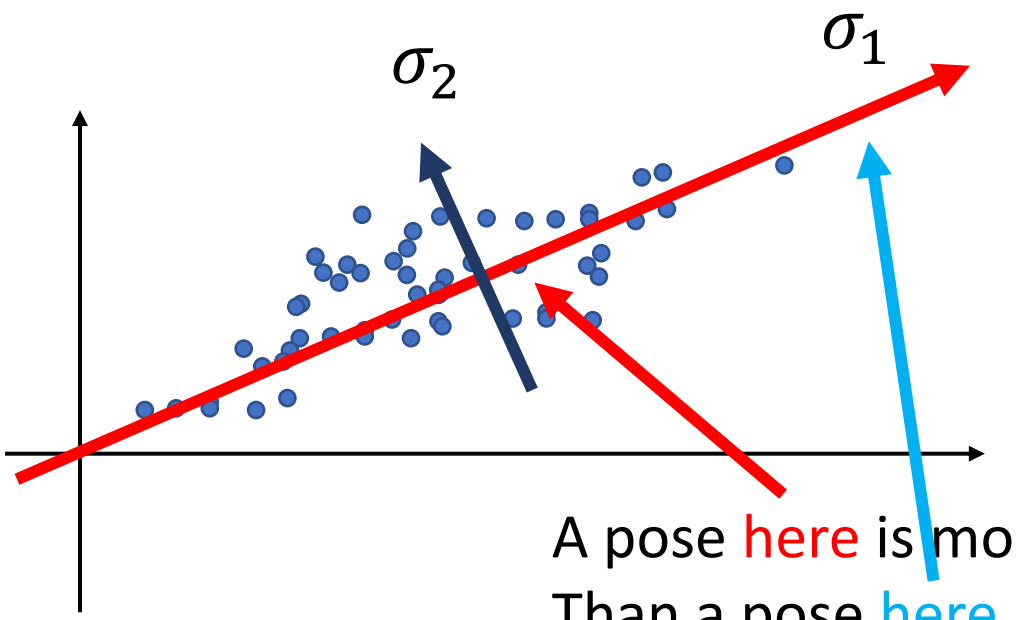
A pose here is more likely to be natural
Than a pose here

PCA Components

$$\boldsymbol{x}_i = \bar{\boldsymbol{x}} + \sum_{k=1}^{n} w_{i,k} \boldsymbol{u}_k$$

a pose $\boldsymbol{x}_i$ with smaller $\sum_k \left(\frac{w_{i,k}}{\sigma_k}\right)^2$

is more likely to be a good pose



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PCA Components

$$\boldsymbol{x} = \bar{\boldsymbol{x}} + \sum_{k=1}^{n} w_k \boldsymbol{u_k}$$

a pose $\boldsymbol{x}$ with smaller $\sum_k \frac{\left((\boldsymbol{x}-\bar{\boldsymbol{x}})\cdot\boldsymbol{u}_k\right)^2}{\sigma_k^2}$

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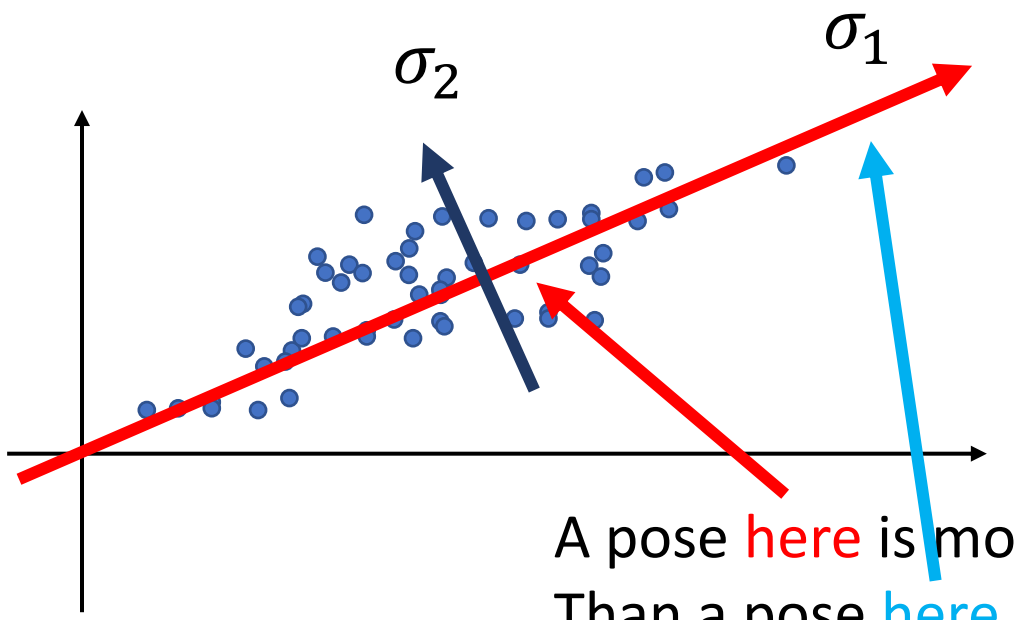

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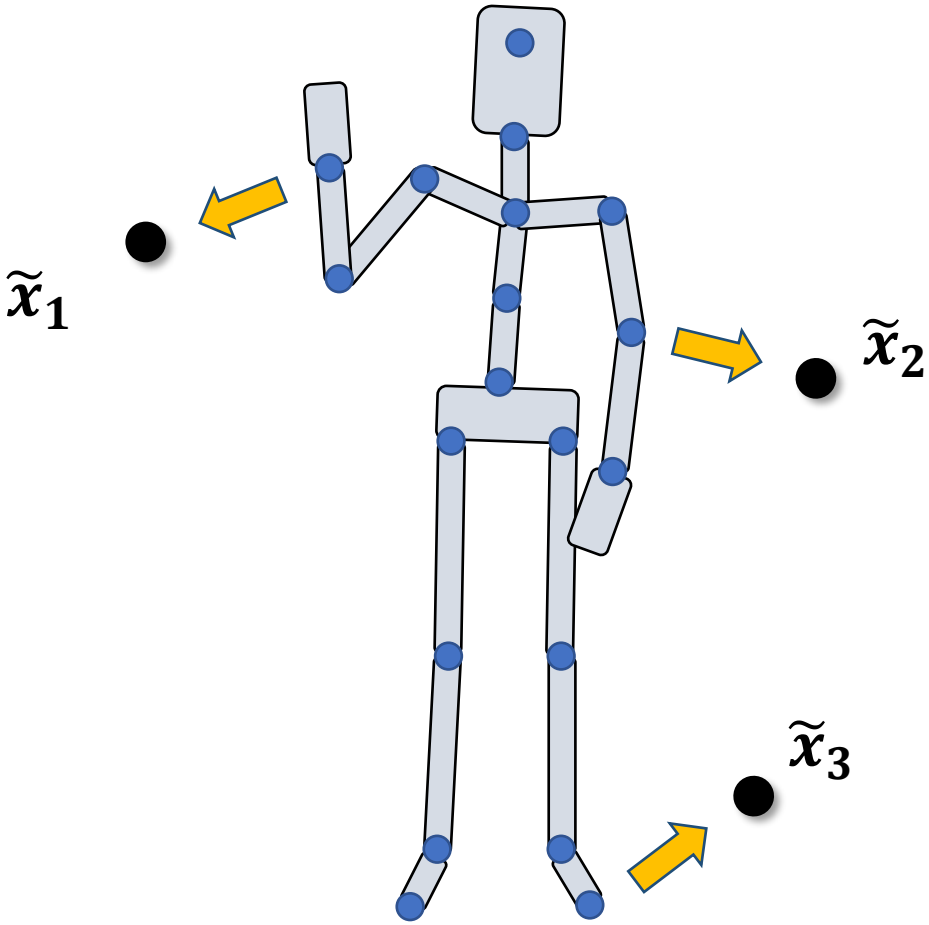
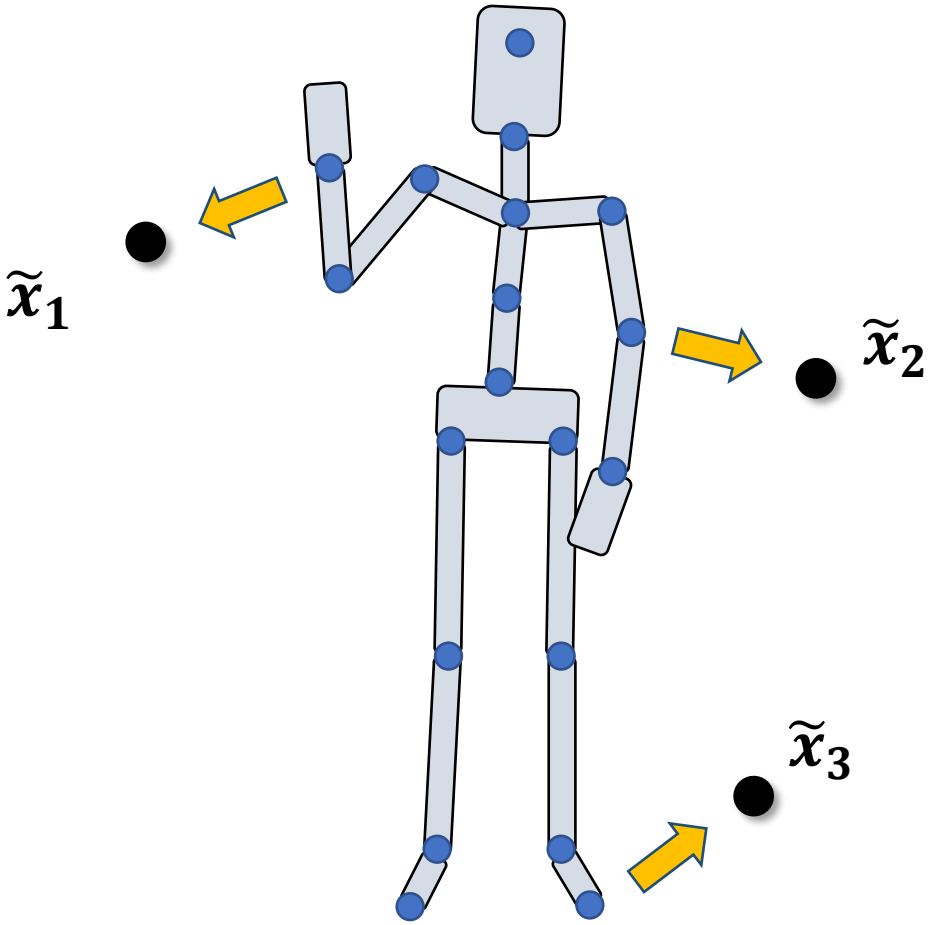
Character IK



$$F(\theta) = \frac{1}{2}\sum_i \|f_i(\boldsymbol{\theta}) - \tilde{\boldsymbol{x}}_i\|_2^2 + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$$

$$\boldsymbol{\theta} = (\boldsymbol{t}_0, R_0, R_1, R_2, \dots\dots)$$

Character IK with a Reference Pose

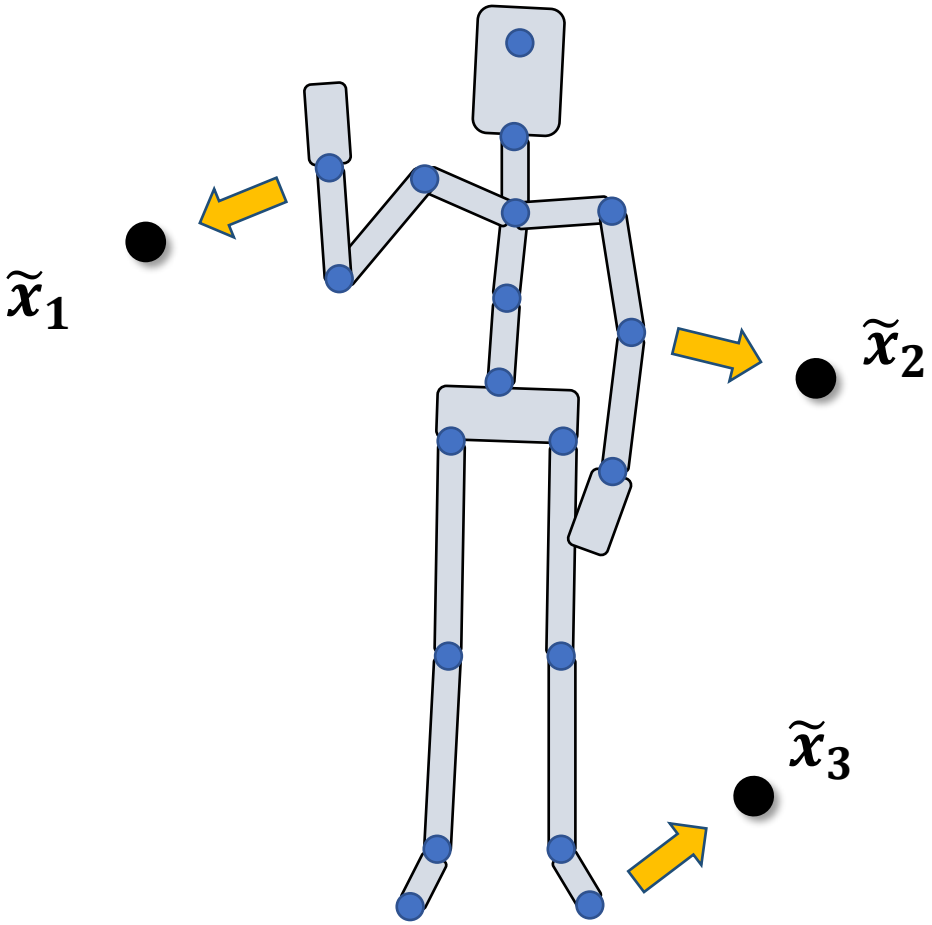


$$F(\theta) = \frac{1}{2}\sum_i \|f_i(\boldsymbol{\theta}) - \tilde{\boldsymbol{x}}_i\|_2^2$$

$$+\frac{\lambda}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|_2^2$$

$$\boldsymbol{\theta} = (\boldsymbol{t}_0, R_0, R_1, R_2, \dots\dots)$$

Character IK with a Motion Prior



$$F(\theta) = \frac{1}{2}\sum_i \|f_i(\boldsymbol{\theta}) - \tilde{\boldsymbol{x}}_i\|_2^2$$

$$+\frac{w}{2}\sum_k \left(\frac{(\boldsymbol{\theta} - \bar{\boldsymbol{\theta}}) \cdot \boldsymbol{u}_k}{\sigma_k}\right)^2$$

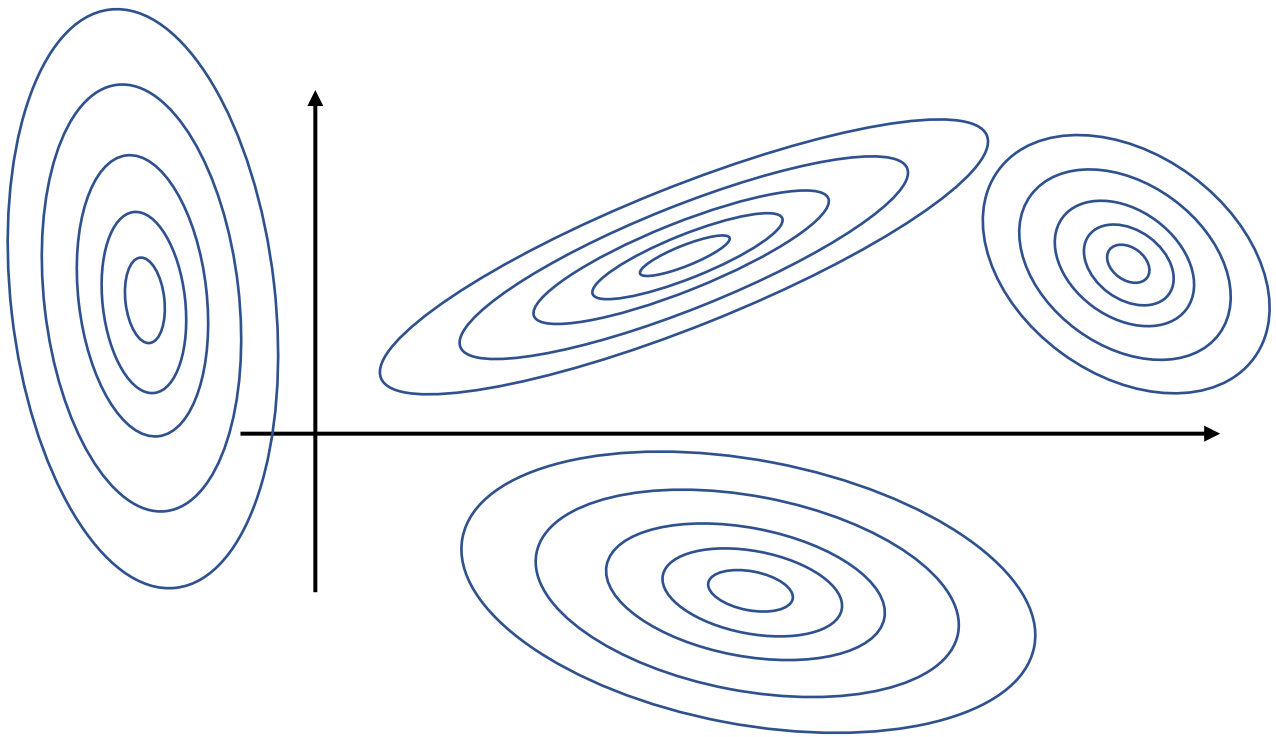
$$\boldsymbol{\theta} = (\boldsymbol{t}_0, R_0, R_1, R_2, \dots\dots)$$

Data Distribution

$p(\boldsymbol{x})$: probability that $\boldsymbol{x}$ is a natural pose

Data Distribution

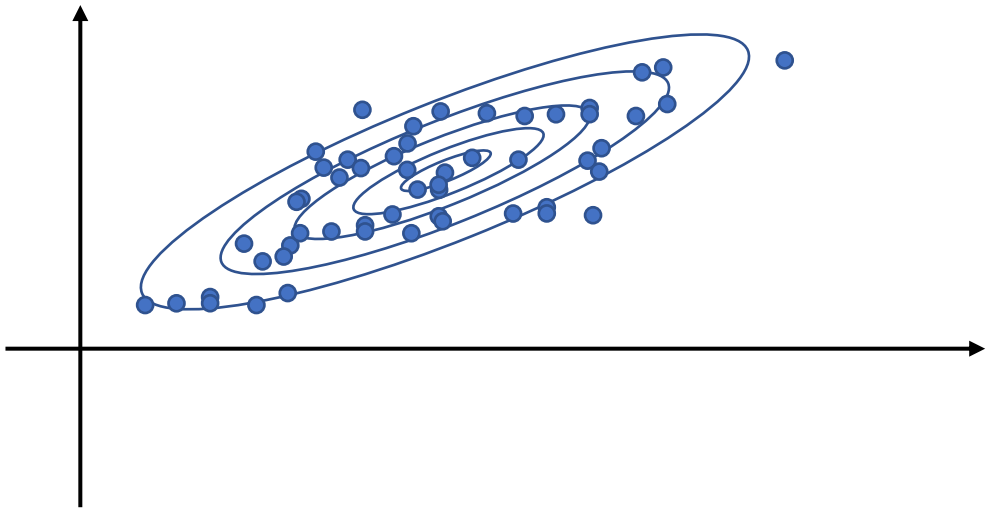
$p(\boldsymbol{x})$: probability that $\boldsymbol{x}$ is a natural pose



Data Distribution

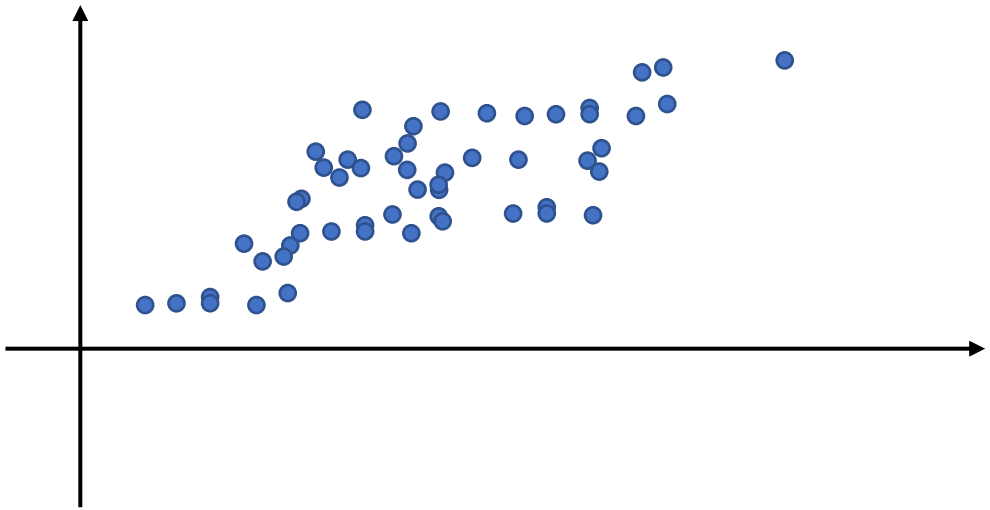
$p(\boldsymbol{x})$: probability that $\boldsymbol{x}$ is a natural pose

a set of data points $\{\boldsymbol{x}_i\} \sim p(\boldsymbol{x})$



Data Distribution

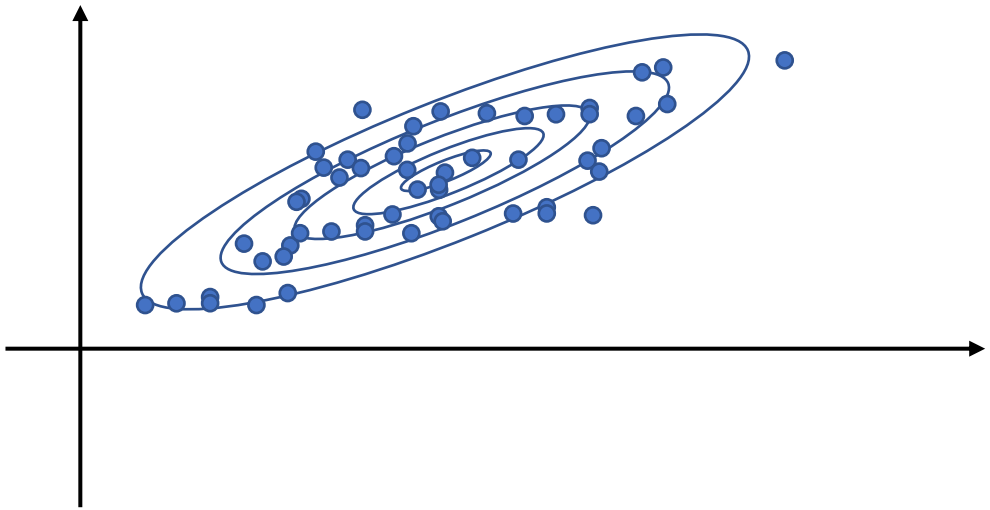
Given a dataset of mocap poses $\{\boldsymbol{x}_i\}$



Data Distribution

Given a dataset of mocap poses $\{\boldsymbol{x}_i\}$

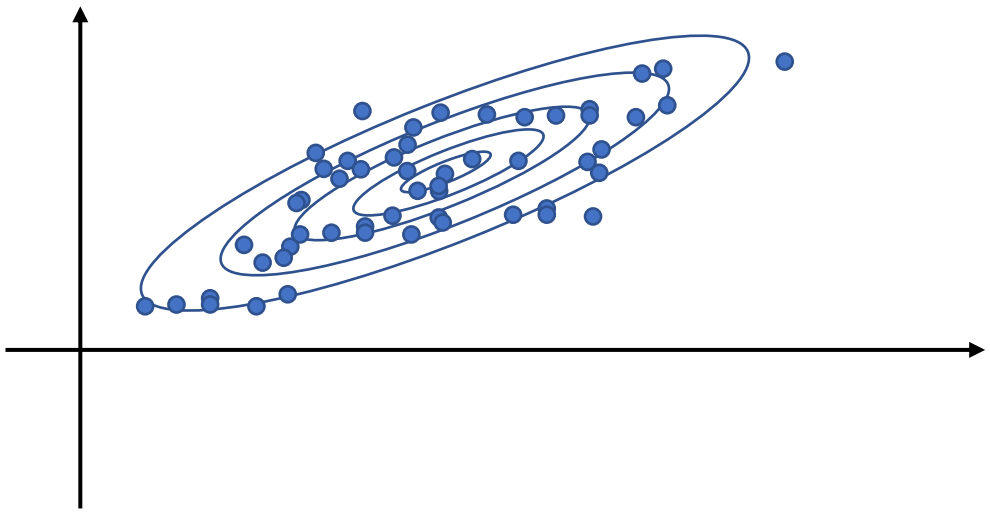
How to find $p(\boldsymbol{x})$?



Gaussian Distribution

Dataset $\{\boldsymbol{x}_i\}$

$$p(\boldsymbol{x}) = \mathcal{N}(\mu_i, \Sigma_i) = \frac{1}{\sqrt{(2\pi)^k|\Sigma|}} e^{-\frac{1}{2}(\boldsymbol{x}-\overline{\boldsymbol{x}})^T\Sigma^{-1}(\boldsymbol{x}-\overline{\boldsymbol{x}})}$$



Gaussian Distribution

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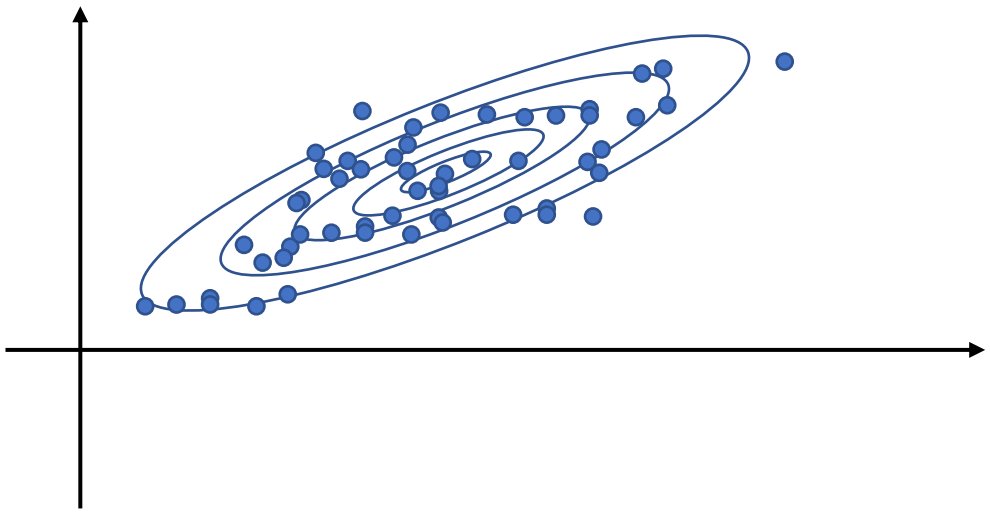


Maximum Likelihood Estimators (MLE):

$$\overline{\boldsymbol{x}} = \frac{1}{N}\sum_i \boldsymbol{x}_i$$

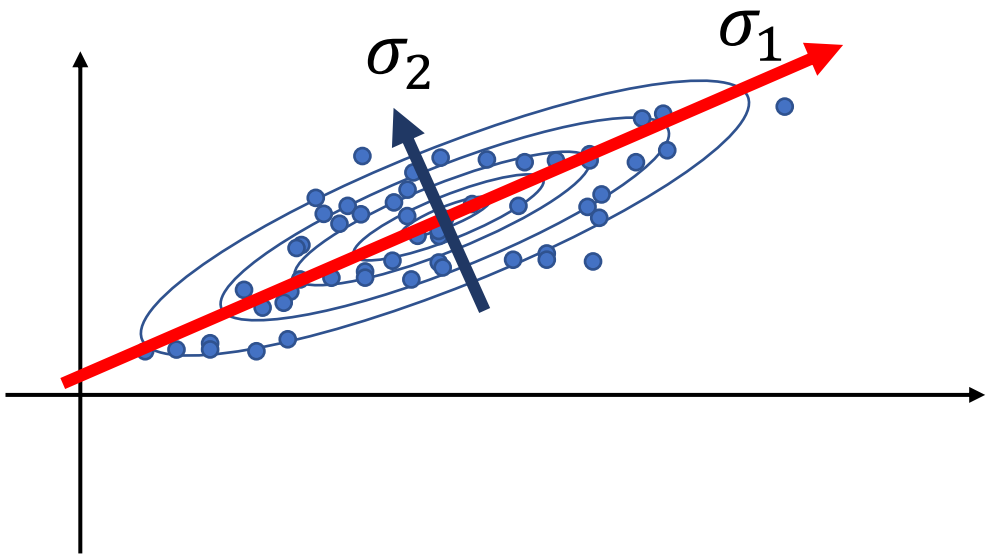
$$\Sigma = \frac{1}{N} X^T X$$

X

PCA and Gaussian Distribution

Dataset $\{\boldsymbol{x}_i\}$

$$p(\boldsymbol{x}) = \mathcal{N}(\mu_i, \Sigma_i) = \frac{1}{\sqrt{(2\pi)^k|\Sigma|}} e^{-\frac{1}{2}(\boldsymbol{x}-\overline{\boldsymbol{x}})^T\Sigma^{-1}(\boldsymbol{x}-\overline{\boldsymbol{x}})}$$



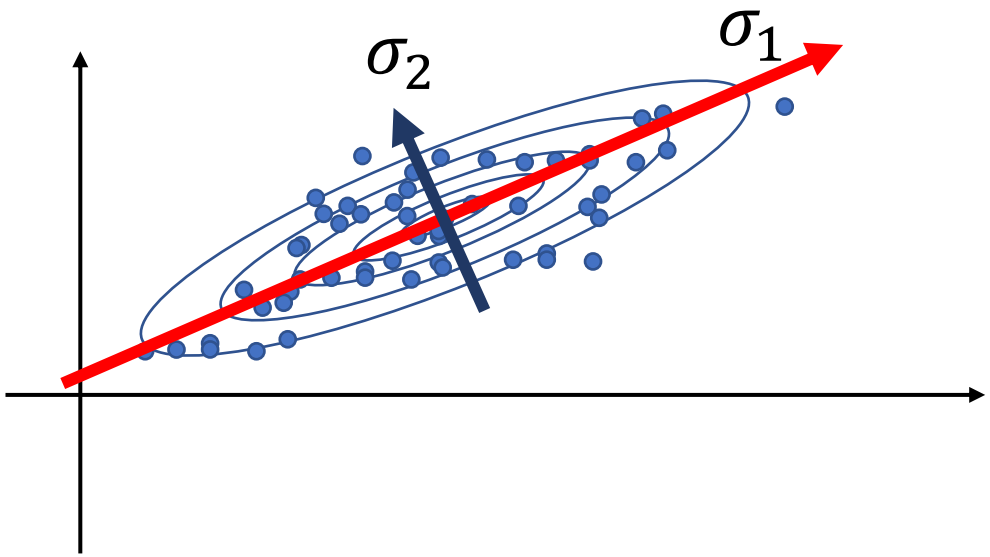
$$\Sigma = X^T X = U \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_N^2 \end{bmatrix} U^T$$

$$\boldsymbol{x} - \overline{\boldsymbol{x}} = \sum_{k=1}^{n} w_k \boldsymbol{u_k}$$

PCA and Gaussian Distribution

Dataset $\{\boldsymbol{x}_i\}$

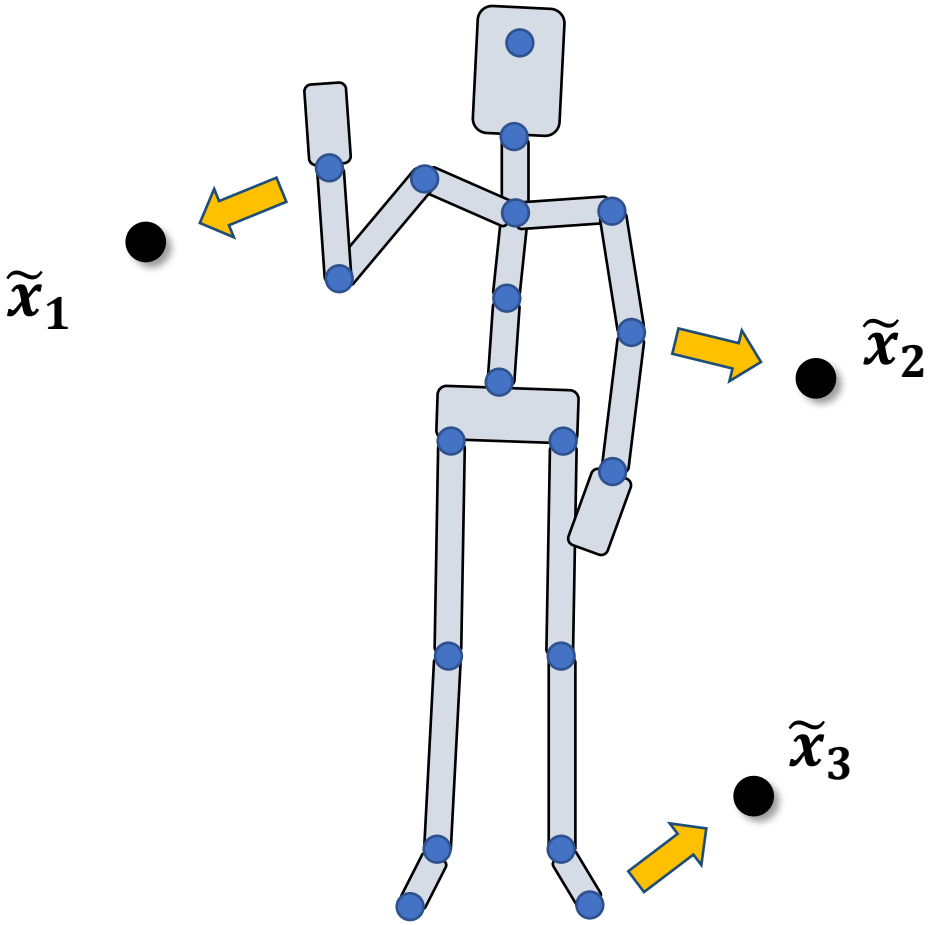
$$p(\boldsymbol{x}) = \mathcal{N}(\mu_i, \Sigma_i) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(\boldsymbol{x}-\overline{\boldsymbol{x}})^T \Sigma^{-1} (\boldsymbol{x}-\overline{\boldsymbol{x}})}$$



$$p(\boldsymbol{x}) = \prod_k \frac{1}{\sigma_k \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{w_k}{\sigma_k}\right)^2}$$

$$w_k = (\boldsymbol{x} - \overline{\boldsymbol{x}}) \cdot \boldsymbol{u}_k$$

Character IK with a Motion Prior

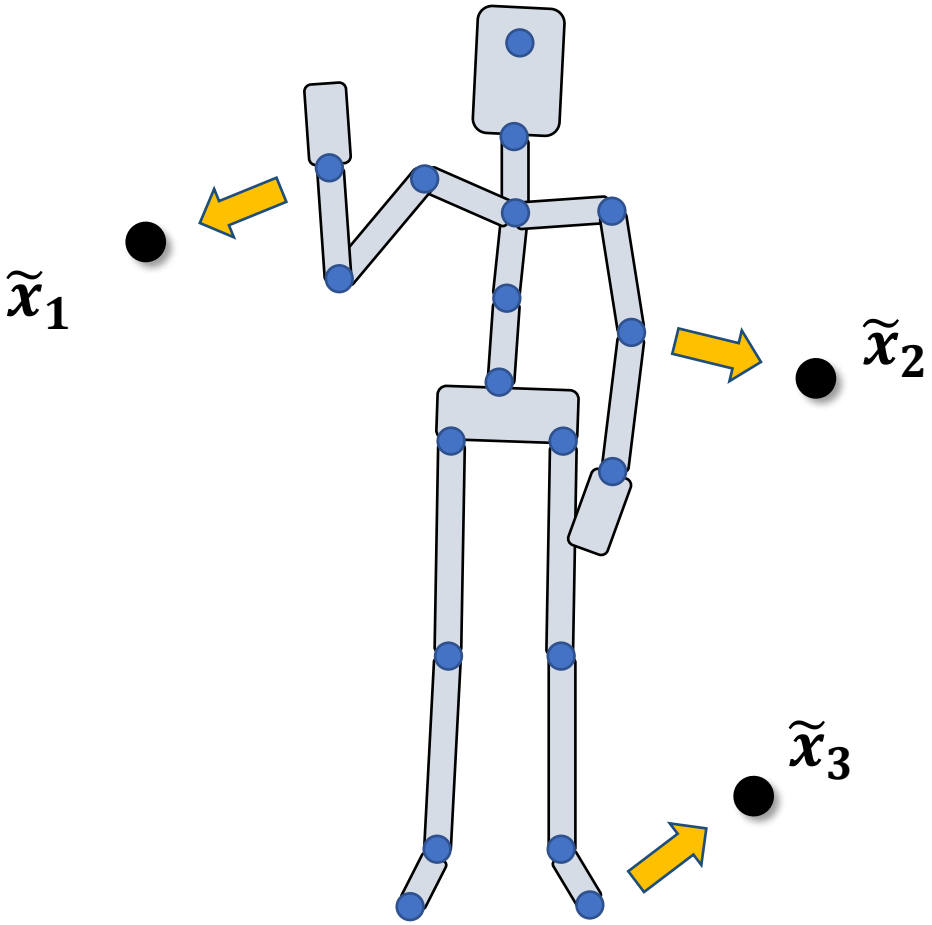


$$F(\theta) = \frac{1}{2}\sum_i \|f_i(\boldsymbol{\theta}) - \tilde{\boldsymbol{x}}_i\|_2^2$$

$$+\frac{w}{2}\sum_k \left(\frac{(\boldsymbol{\theta} - \bar{\boldsymbol{\theta}}) \cdot \boldsymbol{u}_k}{\sigma_k}\right)^2$$

$$\boldsymbol{\theta} = (\boldsymbol{t}_0, R_0, R_1, R_2, \dots\dots)$$

Character IK with a Motion Prior

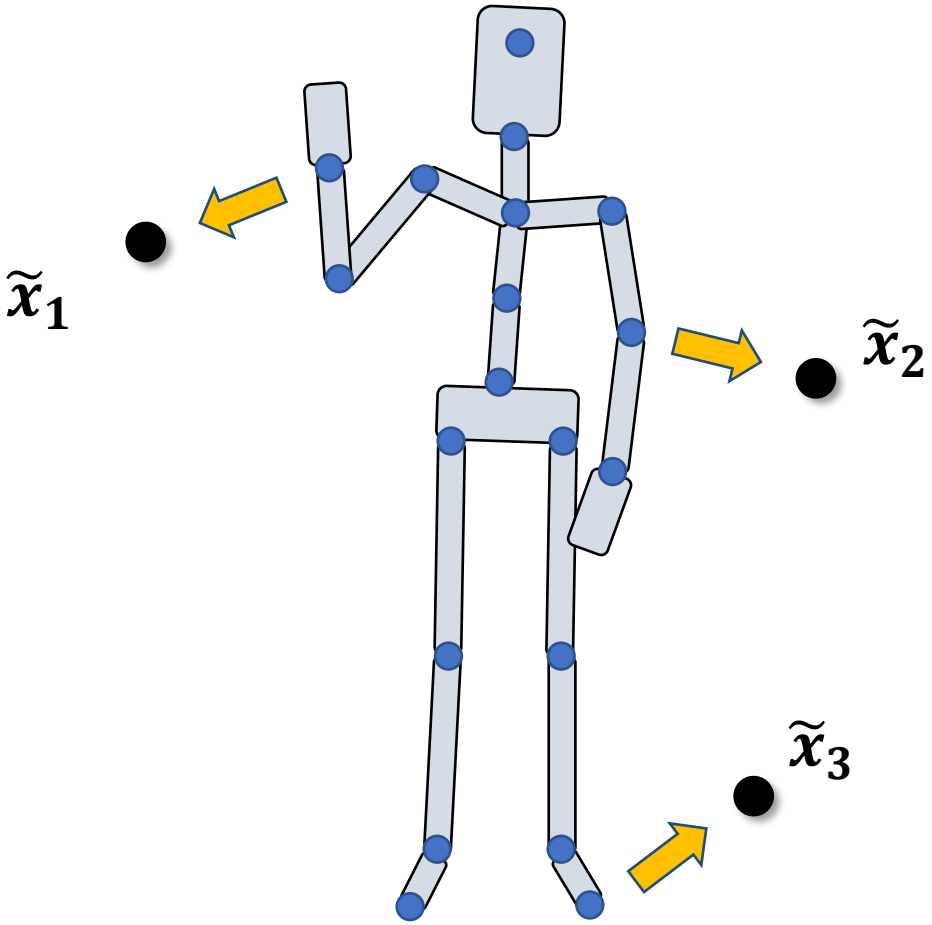


$$F(\theta) = \frac{1}{2}\sum_i \|f_i(\boldsymbol{\theta}) - \tilde{\boldsymbol{x}}_i\|_2^2$$

$$-w \log \prod_k e^{-\frac{1}{2}\left(\frac{(\boldsymbol{\theta}-\bar{\boldsymbol{\theta}})\cdot \boldsymbol{u}_k}{\sigma_k}\right)^2}$$

$$\boldsymbol{\theta} = (\boldsymbol{t}_0, R_0, R_1, R_2, \dots\dots)$$

Character IK with a Motion Prior

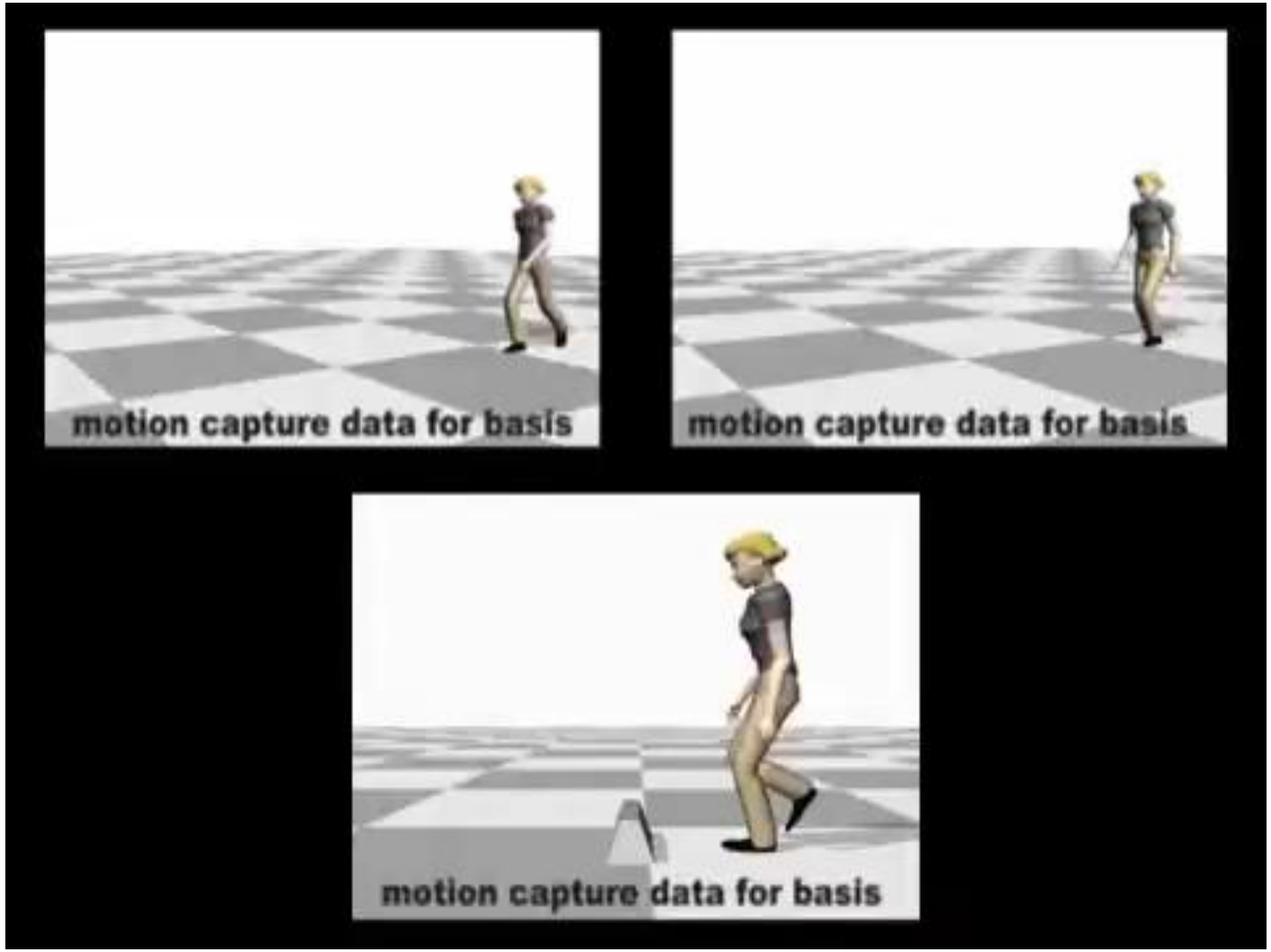


$$F(\theta) = \frac{1}{2}\sum_i \|f_i(\boldsymbol{\theta}) - \tilde{\boldsymbol{x}}_i\|_2^2$$

$$-w \log p(\theta)$$

$$\boldsymbol{\theta} = (\boldsymbol{t}_0, R_0, R_1, R_2, \dots\dots)$$

Motion Synthesis with a Motion Prior



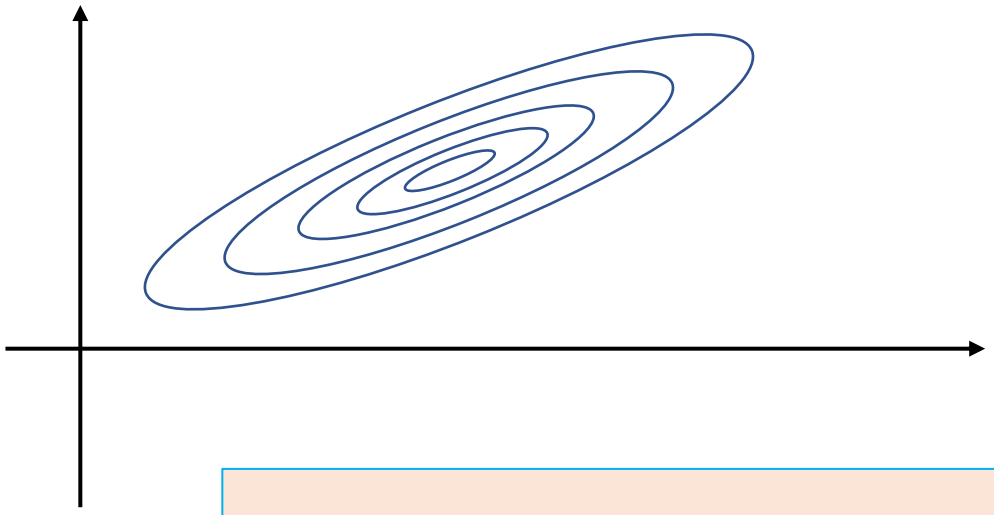
Synthesizing Physically Realistic Human Motion in Low-Dimensional, Behavior-Specific Spaces

Alla Safonova Jessica K. Hodgins Nancy S. Pollard
School of Computer Science
Carnegie Mellon University *

*SIGGRAPH 2004

Gaussian Distribution is not Enough!

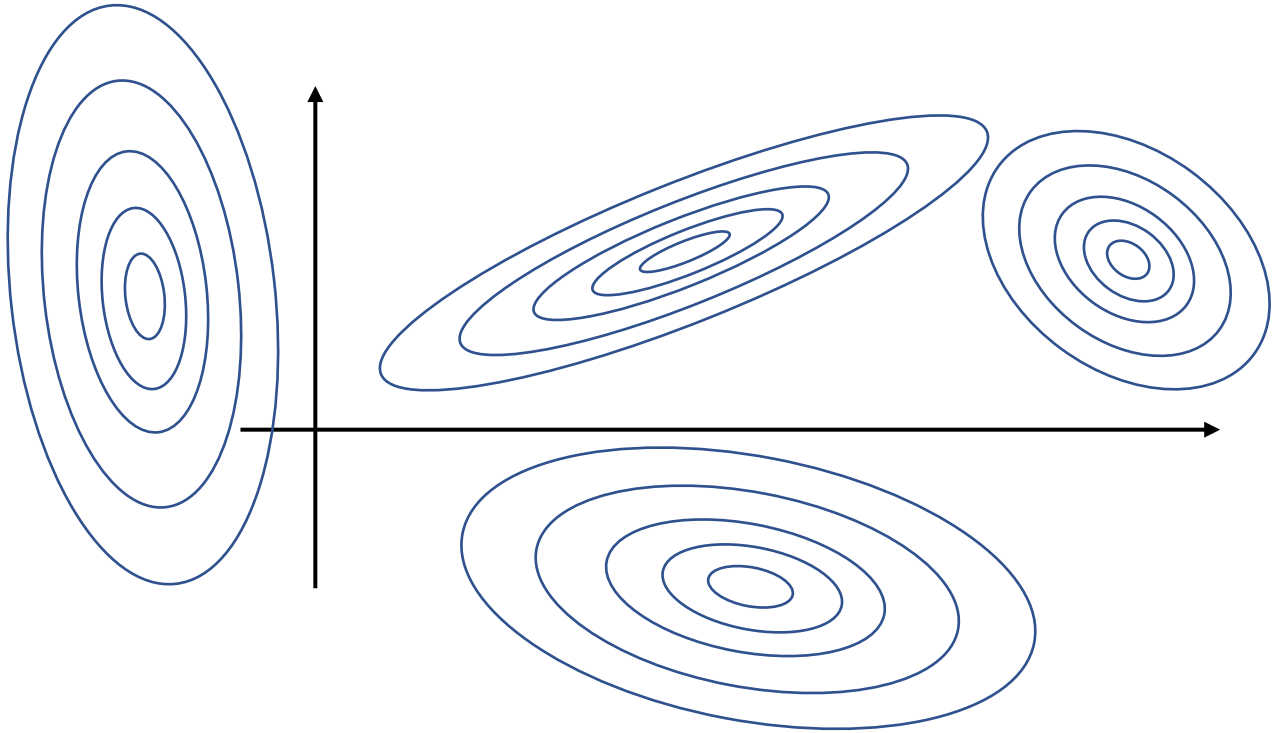
$p(\boldsymbol{x})$: motion prior



$$p(\boldsymbol{x}) = \mathcal{N}(\mu_i, \Sigma_i) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(\boldsymbol{x}-\overline{\boldsymbol{x}})^T \Sigma^{-1} (\boldsymbol{x}-\overline{\boldsymbol{x}})}$$

Gaussian Distribution is not Enough!

$p(\boldsymbol{x})$: motion prior



Gaussian Distribution is not Enough!

$p(\boldsymbol{x})$: motion prior

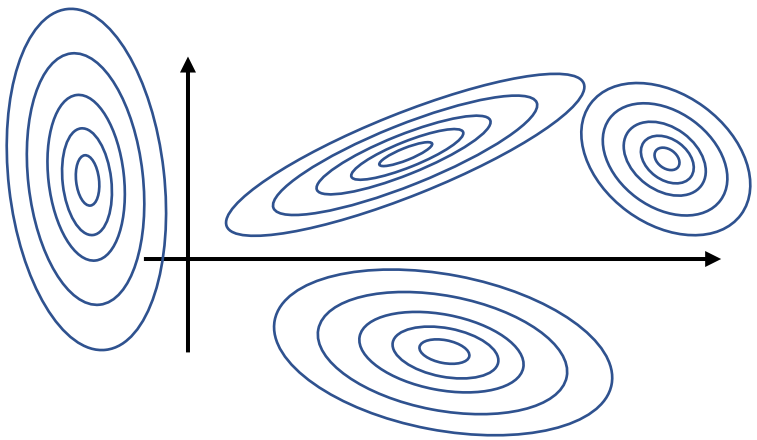
Interactive Generation of Human Animation with Deformable Motion Models

Jianyuan Min
Texas A&M University

Yen-Lin Chen
Texas A&M University

Jinxiang Chai
Texas A&M University

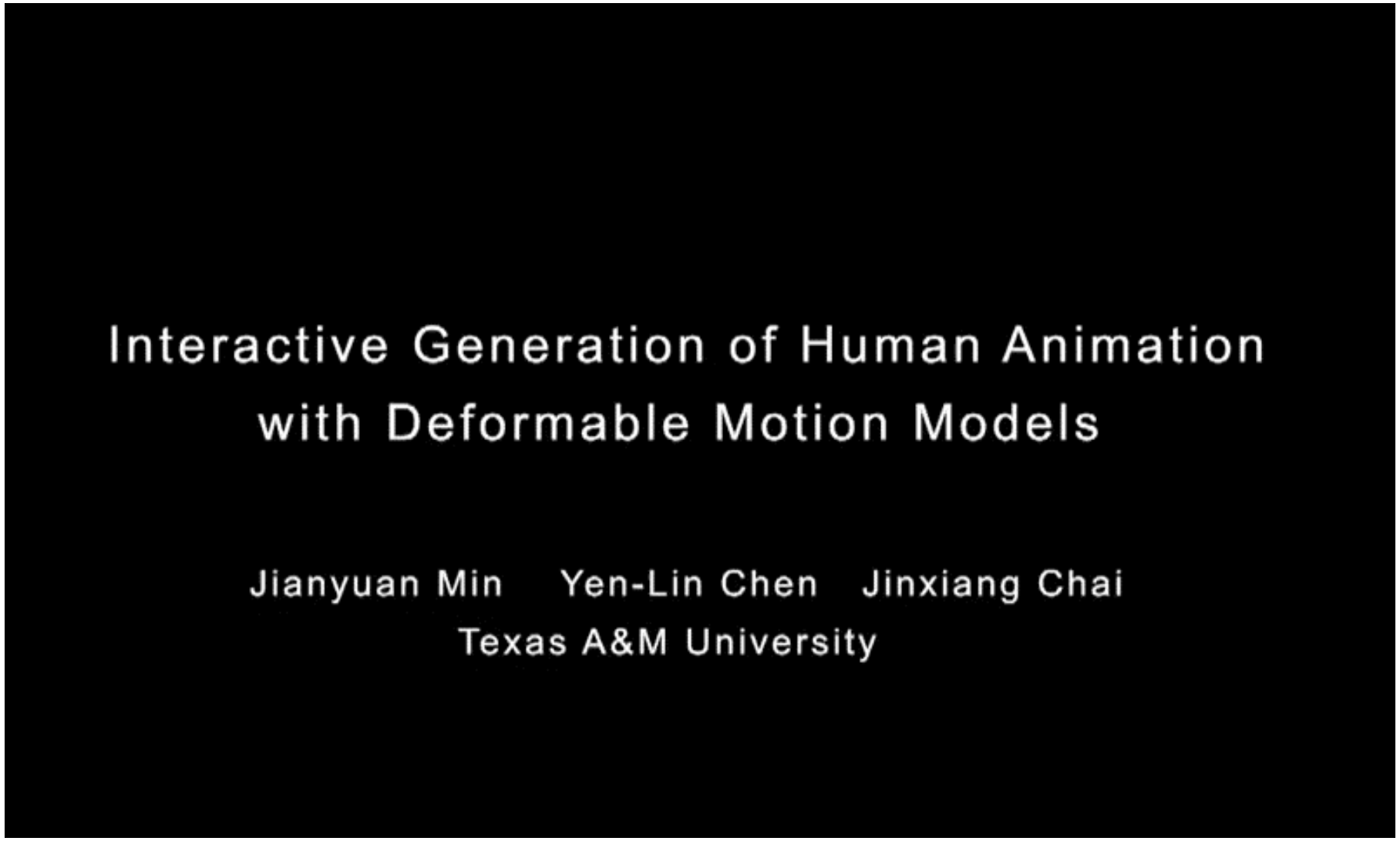
* SIGGRAPH 2009



Gaussian Mixture Models (GMM)

$$p(\boldsymbol{x}) = \sum_i \phi_i \mathcal{N}(\mu_i, \Sigma_i)$$

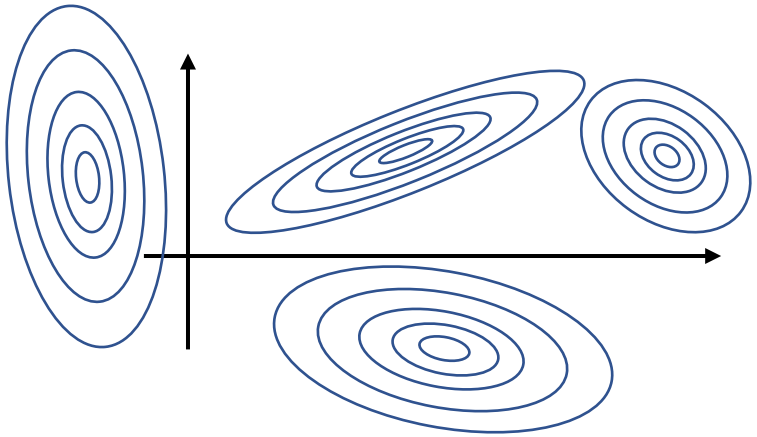
Motion Synthesis with a Motion Prior



Min et al. 2009

Gaussian Distribution is not Enough!

$p(\boldsymbol{x})$: motion prior



Continuous Character Control with Low-Dimensional Embeddings

Sergey Levine[1] Jack M. Wang[1] Alexis Haraux[1] Zoran Popović[2] Vladlen Koltun[1]
[1]Stanford University [2] University of Washington

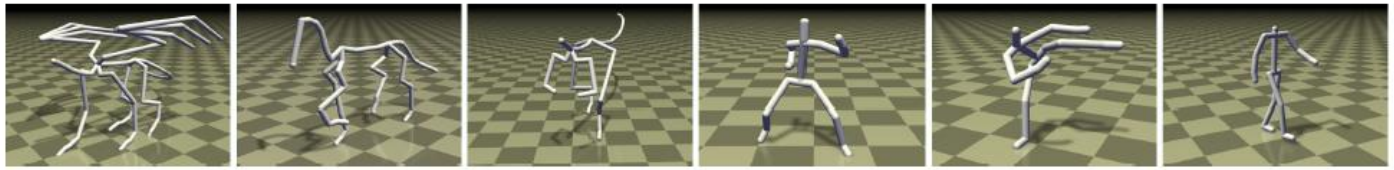
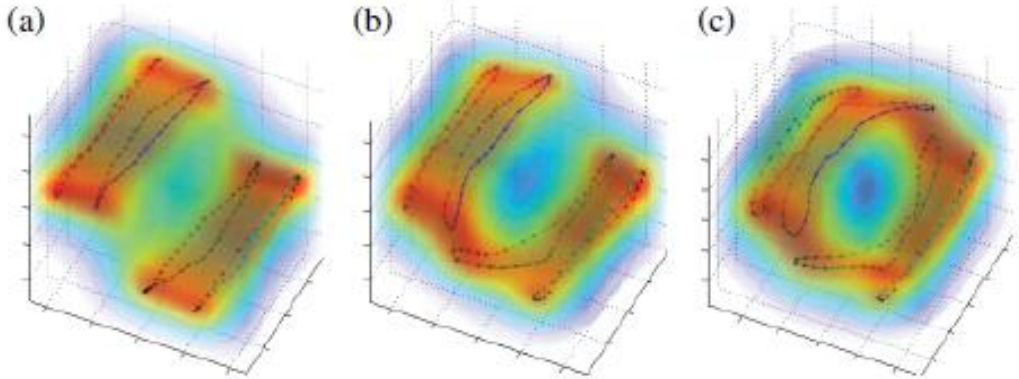


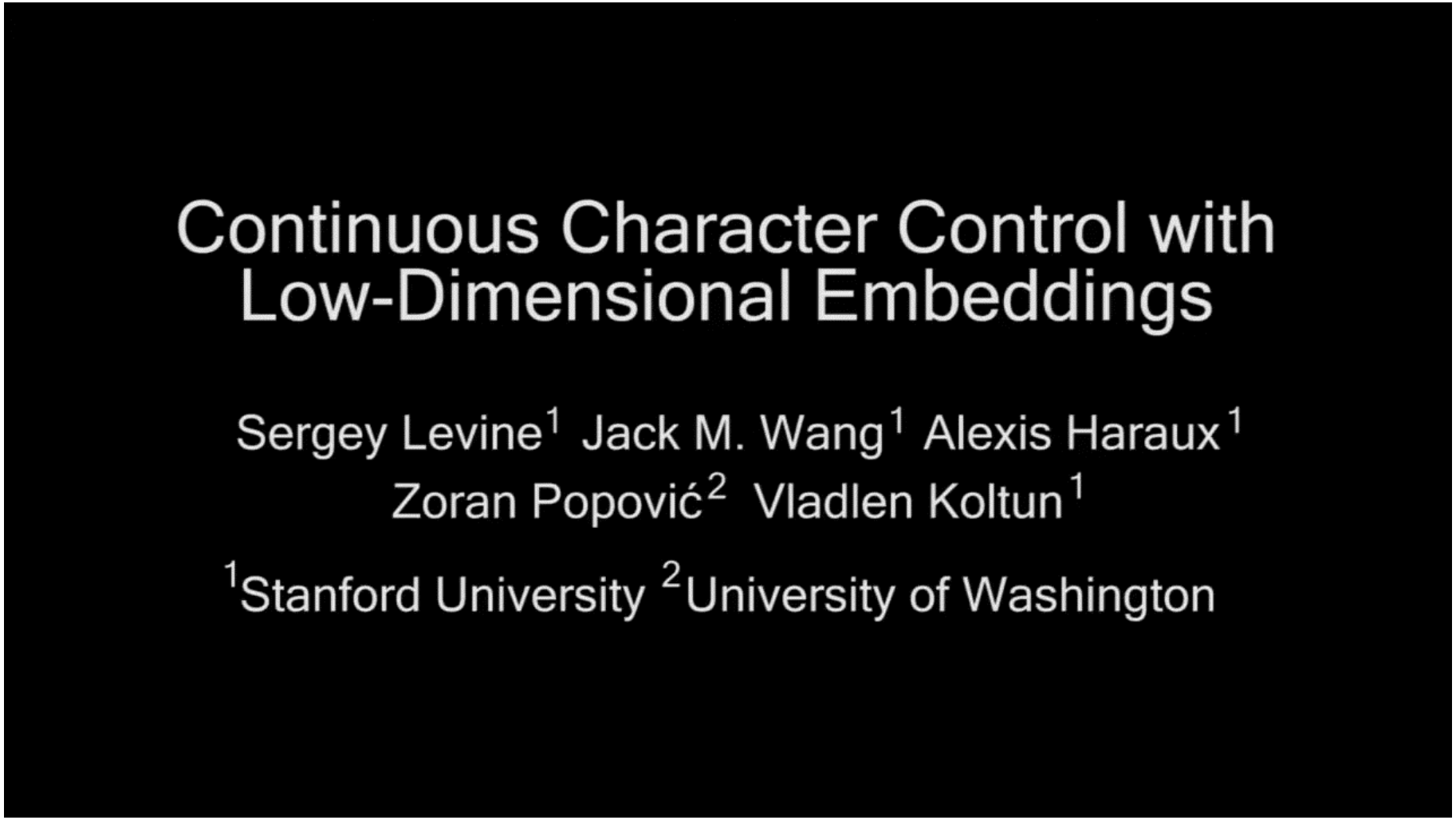
Figure 1: *Character controllers created using our approach: animals, karate punching and kicking, and directional walking.*

* SIGGRAPH 2012

Gaussian Process Latent Variable Model (GPLVM)



Motion Synthesis with a Motion Prior

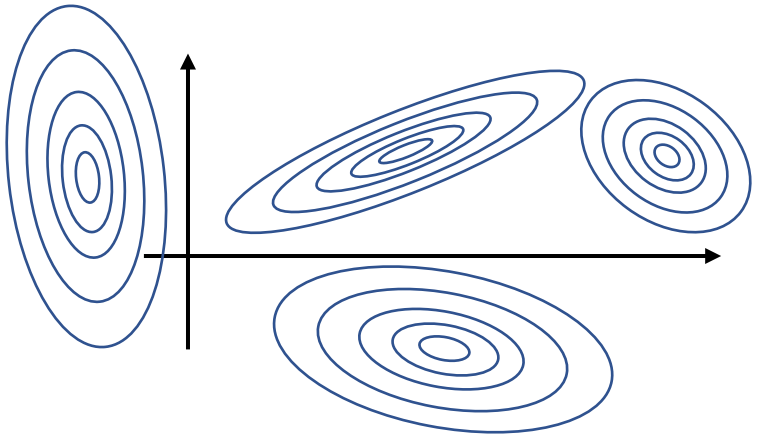


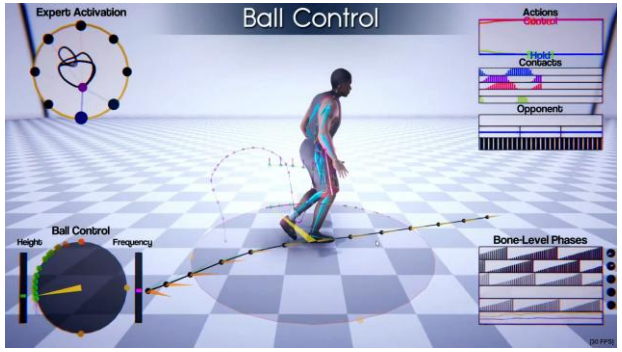
Levine et al. 2012

Gaussian Distribution is not Enough!

Neural networks…

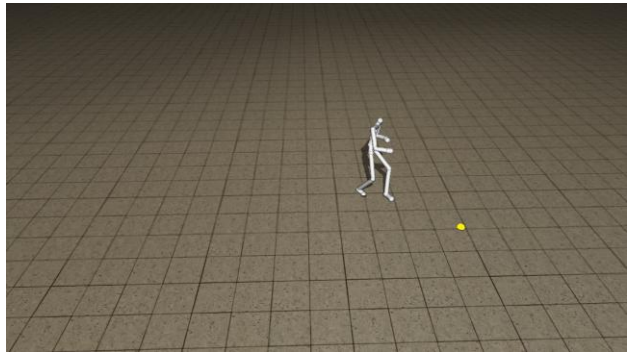
$p(\boldsymbol{x})$: motion prior





[Starke et al 2020, Local Motion Phases for Learning Multi-Contact Character Movements]

[Henter et al. 2020, MoGlow: Probabilistic and Controllable Motion Synthesis Using Normalising Flows]



[Lee et al 2019, Interactive Character Animation by Learning Multi-Objective Control]

[Holden et al 2020, Learned Motion Matching]

Questions?